

Appunti di Analisi Armonica

Appunti dalle Lezioni di J. Bellazini

Moretti Andrea & Rocca Andrea

Aggiornati al: October 9, 2025

Prefazione

Work in progress

Andrea Rocca

Contents

1	The interpolation theory	5
1.1	The Fourier transform	5
1.2	Fourier Transformation for L^p functions, $p \in [1, 2]$	11
1.2.1	$q = p'$	11
1.2.2	Counterexample for $p > 2$	11
1.2.3	The case $p \in [1, 2]$ and the Riesz-Thorin Theorem	12
1.2.4	A generalisation of Riesz-Thorin theorem	18
1.3	Marcinkiewitz theorem	23
2	The Hardy-Littlewood-Sobolev inequality	29
2.1	The maximal function	31
2.1.1	Applications	38
2.2	Hardy-Littlewood-Sobolev inequality	41
2.2.1	Applications	44
3	The Hilbert and Riesz Transform	47
3.1	Schwartz class and Distributions	47
3.1.1	Schwartz class of functions	47
3.1.2	Distributions	54
3.1.3	Sobolev Inequality	57
3.2	The Hilbert and Riesz transform	61
3.2.1	Hilbert Transform	62
3.2.2	Riesz Transform	67
3.3	Calderón-Zygmund's theory	69
3.4	Littlewood-Paley theory	78

Chapter 1

The interpolation theory

1.1 The Fourier transform

In this chapter we will introduce the interpolation theory. One of its consequences will be the definition of the Fourier transform in L^p with $p \in [1, 2]$.

Definition 1.1.1 – Fourier transform in L^1

Given $f \in L^1(\mathbb{R}^d)$, we define the *Fourier transform* of f as the function

$$\hat{f}(\xi) := \int_{\mathbb{R}^d} f(x) e^{-2\pi i \xi \cdot x} dx.$$

Theorem 1.1.2 – Minkowski's theorem

Let us consider $(X, \mu), (Y, \nu)$ σ -finite measurable spaces. Given $f : X \times Y \rightarrow \mathbb{R}$ measurable, if $1 \leq p \leq +\infty$, it holds that

$$\left\| \int_Y f d\nu \right\|_{L_X^p} \leq \int_Y \|f\|_{L_X^p} d\mu.$$

Proof. By duality we have that

$$\|f\|_{L^p} = \sup_{\|g\|_{L^q}=1} |\langle f, g \rangle|.$$

By direct computation we get

$$\begin{aligned} \left| \int_X \int_Y f g d\nu d\mu \right| &\leq \int_Y \int_X |f| |g| d\mu d\nu \\ &\leq \int_Y \|f\|_{L_x^p} \|g\|_{L_x^q} d\nu = \int_Y \|f\|_{L_x^p} d\nu \end{aligned}$$

where in the last inequality we have used Hölder. □

Definition 1.1.3 – Convolution

Given f, g functions on $\mathbb{R}^d$, we define the *convolution* of f and g as the function

$$f * g(x) = \int_{\mathbb{R}^d} f(x - y) g(y) dy.$$

Osservazione 1.1.1

Note that $f * g = g * f$ where the convolution is defined.

In order to define the convolution of two functions one has to require some “regularity” on f, g . One example is given by the following result.

Theorem 1.1.4 – Young's inequality

Given $f \in L^p$ and $g \in L^1$:

$$\|f * g\|_{L^p} \leq \|f\|_{L^p} \|g\|_{L^1}.$$

Proof. By using Minkowski's inequalities we get¹

$$\begin{aligned} \left\| \int f(x-y)g(y)dy \right\|_{L^p} &\leq \int \|f(x-y)g(y)\|_{L^p} dy \\ &= \int |g(y)| \|f(x-y)\|_{L^p} dy = \|f\|_{L^p} \|g\|_{L^1}. \end{aligned}$$

□

Definition 1.1.5 – Approximation of the Identity

A family of functions $\{K_n\}_{n \in \mathbb{N}} \in L^1$ is called an *approximation of the identity* if

- For all n $\int_{\mathbb{R}^d} K_n dx = 1$.
- For all n $\sup_n \int_{\mathbb{R}^d} |K_n| < +\infty$.
- For all $\delta > 0$ $\int_{|x|>\delta} |K_n| dx \xrightarrow{n \rightarrow +\infty} 0$.

Example 1.1.6. Given a function K such that $\int_{\mathbb{R}^d} K = 1$ and $\int_{\mathbb{R}^d} |K| < +\infty$, the family of functions $\{n^d(K(nx))\}_n$ is an approximation of the identity.

Proposition 1.1.7

Let us consider an approximation of the identity $\{K_n\}_n$ and $f \in L^p$ with $1 \leq p < +\infty$. Then

$$K_n * f \longrightarrow f \text{ in } L^p.$$

Proof. Our goal is to show that

$$\|K_n * f - f\|_{L^p} \xrightarrow{n \rightarrow \infty} 0$$

¹In the last equality we use $\|f(x-h)\|_{L^p} = \|f(x)\|_{L^p} \forall h \in \mathbb{R}$, that holds since we are working on an Euclidian space.

By direct computation we get

$$\begin{aligned}
 \left\| \int_{\mathbb{R}^d} K_n(x-y)f(y)dy - f(x) \cdot 1 \right\|_{L^p} &= \left\| \int_{\mathbb{R}^d} K_n(y)f(x-y)dy - f(x) \int_{\mathbb{R}^d} K_n(y)dy \right\|_{L^p} \\
 &= \left\| \int_{\mathbb{R}^d} K_n(y) (f(x-y) - f(x)) dy \right\|_{L^p} \\
 &\leq \int_{\mathbb{R}^d} |K_n(y)| \left\| (f(x-y) - f(x)) \right\|_{L^p} dy
 \end{aligned}$$

Idea The trick here is to fix a $\delta > 0$ (that we will decide later) and split the domain in $\mathbb{R}^n = \{|y| > \delta\} \sqcup \{|y| < \delta\}$. By doing that we can study the two integrals separately. $\square$

The first term becomes:

$$\int_{\{|y| > \delta\}} |K_n(y)| \left\| (f(x-y) - f(x)) \right\| dy \leq 2\|f\| \int_{\{|y| > \delta\}} |K_n(y)| dy \longrightarrow 0$$

The second term requires the continuity of the L^p norm, for finite² p , under the action of shifts. We shall remember that

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that } |y| < \delta \implies \|f(x-y) - f(x)\|_{L^p} \leq \epsilon \quad (1.1)$$

By fixing $\epsilon > 0$ and adjusting δ as in (1.1) we get

$$\int_{\{|y| < \delta\}} |K_n(y)| \left\| (f(x-y) - f(x)) \right\| dy \leq \epsilon \sup_n \int_{\{|y| > \delta\}} |K_n(y)| dy \leq C\epsilon.$$

By letting $\epsilon \longrightarrow 0$ the proof is completed. $\square$

Proposition 1.1.8

We recall some basic fact about the Fourier transform, where we denote $\tau_h f(x) = f(x-h)$ and $\delta_\lambda f(x) = f\left(\frac{x}{\lambda}\right)$.

1. $f \rightarrow \hat{f}$ is linear.
2. $\widehat{\tau_h f}(\xi) = e^{-2\pi i \xi \cdot h} \hat{f}$.
3. $\widehat{\delta_\lambda f}(\xi) = \lambda^d \hat{f}(\lambda \xi)$.
4. If $f, g \in L^1$ then $\widehat{f * g} = \hat{f} \hat{g}$.
5. The function $g_\lambda(x) = e^{-\pi \lambda |x|^2}$ is such that $\hat{g}_\lambda(\xi) = \lambda^{-\frac{d}{2}} e^{-\frac{\pi |\xi|^2}{\lambda}}$.

Proof. These are consequences of some direct computations. We do explicitly the last one:

²This propriety is not true for $p = \infty$.

for the dilatation formula it is sufficient to prove it for the case $\lambda = 1$:

$$\begin{aligned}\hat{g}(\xi) &= \int_{\mathbb{R}^d} e^{-\pi|x|^2} e^{-2\pi i\xi \cdot x} dx \\ &= \int_{\mathbb{R}^d} e^{-\pi(x+i\xi)^2} e^{-\pi\xi^2} dx \\ &= e^{-\pi\xi^2} \underbrace{\int_{\mathbb{R}^d} e^{-\pi(x+i\xi)^2} dx}_{h(\xi) \equiv 1} \\ &= e^{-\pi\xi^2}\end{aligned}$$

indeed we have that

$$h'(\xi) = \int_{\mathbb{R}^d} e^{-\pi(x+i\xi)^2} (-2\pi i(x+i\xi)) dx = i \int_{\mathbb{R}^d} \frac{d}{dx} [e^{-\pi(x+i\xi)^2}] dx = 0.$$

and $h(0) = 1$. □

Theorem 1.1.9 – Plancherel

If we have $f \in L^1 \cap L^2$ then $\hat{f} \in L^2$ and $\|f\|_{L^2} = \|\hat{f}\|_{L^2}$.

Proof. Thanks to dominated convergence we know that

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)|^2 e^{-\epsilon\pi|\xi|^2} d\xi \xrightarrow{\epsilon \rightarrow +\infty} \int_{\mathbb{R}^d} |\hat{f}(\xi)|^2 d\xi.$$

and from the uniqueness of the limit the proof is completed if we show that

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)|^2 e^{-\epsilon\pi|\xi|^2} d\xi \xrightarrow{\epsilon \rightarrow +\infty} \int_{\mathbb{R}^d} |f(x)|^2 dx.$$

By direct computation we have

$$\begin{aligned}\int_{\mathbb{R}^d} |\hat{f}(\xi)|^2 e^{-\epsilon\pi|\xi|^2} d\xi &= \int_{\mathbb{R}^d} \hat{f}(\xi) \overline{\hat{f}(\xi)} e^{-\epsilon\pi|\xi|^2} d\xi \\ &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx \right) \left(\int_{\mathbb{R}^d} \overline{f(y)} e^{+2\pi i y \cdot \xi} dy \right) e^{-\epsilon\pi|\xi|^2} d\xi \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x) \overline{f(y)} \left(\int_{\mathbb{R}^d} e^{-2\pi i(x-y) \cdot \xi} e^{-\epsilon\pi|\xi|^2} d\xi \right) dx dy.\end{aligned}$$

Since the expression between brackets is the Fourier transformation of the rescaled Gaussian we get

$$\int_{\mathbb{R}^d} e^{-2\pi i(x-y) \cdot \xi} e^{-\epsilon\pi|\xi|^2} d\xi = \epsilon^{-\frac{d}{2}} e^{-\frac{\pi}{\epsilon}(x-y)^2}$$

and by substituting back

$$\begin{aligned}\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x) \overline{f(y)} \left(\int_{\mathbb{R}^d} e^{-2\pi i(x-y) \cdot \xi} e^{-\epsilon\pi|\xi|^2} d\xi \right) dx dy &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x) \overline{f(y)} \epsilon^{-\frac{d}{2}} e^{-\frac{\pi}{\epsilon}(x-y)^2} dx dy \\ &= \int_{\mathbb{R}^d} f(x) \int_{\mathbb{R}^d} \overline{f(y)} \epsilon^{-\frac{d}{2}} e^{-\frac{\pi}{\epsilon}(x-y)^2} dy dx \\ &= \int_{\mathbb{R}^d} f(x) (K_\epsilon * \bar{f}) dx\end{aligned}$$

where $K_\epsilon = \epsilon^{-\frac{d}{2}} e^{-\frac{\pi}{\epsilon}(x-y)^2}$ is an approximation of the identity. Since

$$K_\epsilon * \bar{f} \xrightarrow{L^2} \bar{f}$$

we have that

$$\int_{\mathbb{R}^d} f(x) (K_\epsilon * \bar{f}) dx \longrightarrow \|f\|_{L^2}^2.$$

□

Corollary 1.1.10 – Plancherel-Parseval Identity

For $f, g \in L^2$ we have

$$\int_{\mathbb{R}^d} f \bar{g} dx = \int_{\mathbb{R}^d} \hat{f} \overline{\hat{g}} dx.$$

Proof. The statement is a simple consequences of the parallelogram law for scalar products:

$$\langle f, g \rangle = \frac{1}{2} \left[\|f + g\|_{L^2}^2 - \|f - g\|_{L^2}^2 \right]$$

□

Idea From Plancherel's Theorem it follows that the map $f \xrightarrow{\mathcal{F}} \hat{f}$ has a unique extension to L^2 and is a surjective isometry. $\lrcorner$

Definition 1.1.11 – Fourier transform in L^2

Given^a $\{f_n\} \in L^1 \cap L^2$ such that $f_n \rightarrow f$ in L^2 , we define the Fourier transform of f as

$$\hat{f} := \lim_{n \rightarrow \infty} \hat{f}_n.$$

^aa sequence exists because $L^1 \cap L^2$ is dense in L^2 .

Osservazione 1.1.2

The limit surely exists because $\{\hat{f}_n\}$ is a Cauchy sequence in L^2 and, thanks to Plancherel-Parseval identity, also $\{f_n\}$ is.

Definition 1.1.12 – Fourier antitransform

Given $f \in L^2$, we define the *Fourier antitransform* as

$$\check{f}(x) = f^\vee(x) := \int_{\mathbb{R}^d} f(\xi) e^{2\pi i \xi \cdot x} d\xi.$$

Osservazione 1.1.3

Since $\check{f}(x) = \hat{f}(-x)$, $\check{f}$ is well-defined by Plancherel's Theorem.

Theorem 1.1.13 – Inversion formula

If $f \in L^2$, then $f = (\hat{f})^\vee$ almost everywhere.

Proof. For $f \in L^2$ the following formula holds

$$\int_{\mathbb{R}^d} \hat{g}_\lambda(x-y) f(y) dy = \int_{\mathbb{R}^d} g_\lambda(\xi) \hat{f}(\xi) e^{2\pi i(\xi-x) \cdot \lambda} d\xi \quad (1.2)$$

where g is the standard gaussian and g_λ is the rescaled one. The proof of (1.2) can be done first by taking $f \in L^1 \cap L^2$, where one can write:

$$\begin{aligned} \int_{\mathbb{R}^d} \hat{g}_\lambda(x-y)f(y)dy &= \int_{\mathbb{R}^d} \hat{g}_\lambda(y-x)f(y)dy \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g_\lambda(\xi) e^{-2\pi i \xi \cdot (y-x)} f(y) d\xi dy \\ &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} f(y) e^{-2\pi i \xi \cdot y} dy \right) g_\lambda(\xi) e^{2\pi i \xi \cdot x} d\xi \\ &= \int_{\mathbb{R}^d} g_\lambda(\xi) \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi. \end{aligned}$$

If $f \in L^2 \setminus L^1$ we can approximate it with $\{f_j\}_{j \in \mathbb{N}} \subset L^1 \cap L^2$ and since

$$\int_{\mathbb{R}^d} \hat{g}_\lambda(x-y) \underbrace{f_j(y)}_{\rightarrow f(y)} dy = \int_{\mathbb{R}^d} g_\lambda(\xi) \underbrace{\hat{f}_j(\xi)}_{\rightarrow \hat{f}(\xi)} e^{2\pi i (\xi-x) \cdot} d\xi$$

and $g_\lambda, \hat{g}_\lambda \in L^2$, the formula holds for f too.

As $\lambda \rightarrow 0$ the left side of (1.2) tends to $f(x)$ in L^2 (it is an approximation of the identity). The right side instead tends to $(\hat{f})^\vee$, indeed

$$g_\lambda \hat{f} \rightarrow \hat{f}$$

in L^2 by dominated convergence and

$$(g_\lambda \hat{f})^\vee \rightarrow (\hat{f})^\vee$$

in L^2 thanks to Plancherel identity. □

1.2 Fourier Transformation for L^p functions, $p \in [1, 2]$

In the last sections we defined the Fourier transformation for functions in L^1 and L^2 . It is natural to ask if it can be defined for functions in L^p with some $p \in [1, \infty]$. The idea is to follow step by step what we have done for $p = 2$ and define $\hat{f}$ for a function $f \in L^p$ as

$$\hat{f} = \lim_{n \rightarrow \infty} \hat{f}_n$$

where $\{f_n\} \in L^1 \cap L^p$ and $f_n \rightarrow f$ in L^p .

Next step is to check whether or not this definition is well-posed (Does $\hat{f}$ stay in L^q for some q ?). In order to do that we will search for an inequality of the type

$$\|\hat{f}\|_{L^q} \leq \|f\|_{L^p} \quad (1.3)$$

for some $p, q \in [1, \infty]$. At the end of the next pages we are going to see that (1.3) holds only for $p \in [1, 2]$ and $q = p'$, this result is called *Hausdorff-Young inequality*.

Therefore it will be possible to define the Fourier transform only for functions in L^p for $p \in [1, 2]$.

1.2.1 $q = p'$

Idea The idea is that if (1.3) hold for a function f then it must hold even for every $f_\lambda(x) = f(\lambda x)$. This allows us to give restriction on the value of q given p , indeed the rescaling of the norm on LHS and RHS have to be the same! Otherwise taking $\lambda \rightarrow 0$ or $\lambda \rightarrow \infty$ the inequality fails. $\lrcorner$

Thanks to this idea we see that

$$\|\hat{f}_\lambda\|_{L^q} = \lambda^{-d} \lambda^{\frac{d}{q}} \|\hat{f}\|_{L^q}$$

and

$$\|f_\lambda\|_{L^p} = \lambda^{-\frac{d}{p}} \|f\|_{L^p}$$

then

$$-d + \frac{d}{q} = -\frac{d}{p}$$

that give us $q = p'$.

1.2.2 Counterexample for $p > 2$

We are going to construct a sequence $\{f_j\}$ of functions in L^p , $p > 2$, such that

$$\frac{\|\hat{f}_j\|_{L^{p'}}}{\|f_j\|_{L^p}} \rightarrow +\infty.$$

This proves that (1.3) is false for every $p > 2$.

Let us define

$$f_N(x) = \sum_{k=1}^N e^{2\pi i x \cdot k v} g(x - n v)$$

where g is the Gaussian and $v \in \mathbb{R}^d$ is a vector with integer modulus.

Idea The idea here is to consider a sequence of functions such that $|\hat{f}_N| = |\bar{f}|$ and $\|f_N\|_{L^p} \sim N^{\frac{1}{p}}$. By doing that

$$\frac{\|\hat{f}_N\|_{L^{p'}}}{\|f_N\|_{L^p}} = \frac{\|f_N\|_{L^{p'}}}{\|f_N\|_{L^p}} \sim N^{\frac{1}{p'} - \frac{1}{p}} \longrightarrow \infty$$

if $p > 2$. ┘

All the calculations require a bit of work, we invite you to try it! (or maybe you can just trust us)

1.2.3 The case $p \in [1, 2]$ and the Riesz-Thorin Theorem

In order to prove (1.3) for $p \in [1, 2]$ we need our first result in interpolation theory. As said before, this result will lead us to the Fourier transform in L^p , with $1 \leq p \leq 2$.

In order to prove the main theorem we first need to develop some results of complex analysis³.

Theorem 1.2.1 – Hadamard Three lines theorem

Let us consider the strip $\Sigma = \{z \in \mathbb{C} : 0 \leq \Re(z) \leq 1\}$ in the complex plane. Given F such that:

- F is analytic on $\overset{\circ}{\Sigma}$
- F is bounded and continuous on Σ
- There exists M_0, M_1 such that
$$\begin{cases} |F(it)| \leq M_0 \\ |F(1+it)| \leq M_1 \end{cases} \quad \text{for all } t \in \mathbb{R}$$

It follows that:

$$|F(z)| \leq M_0^{1-t} M_1^t$$

where $t = \Re(z)$.

Idea This result allows us to control F inside the strip, knowing only a control on its boundary. ┘

Proof. First we assume $M_0 = M_1 = 1$. If this is not the case we can consider the function

$$\tilde{F}(z) = \frac{F(z)}{M_0^{1-z} M_1^z}$$

which still satisfies the hypothesis of the theorem and $\tilde{M}_0 = \tilde{M}_1 = 1$. Once we prove the theorem for $\tilde{F}$ is trivial to deduce it for F .

Now, the theorem is proved if we show $|F(z)| \leq 1$ for every $z \in \Sigma$. We define

$$\tilde{g}(y) := \sup_{0 \leq x \leq 1} |F(x + iy)|$$

for every $y \in \mathbb{R}$, which is the upper bound of the function F on a fixed horizontal segment in Σ . By looking at $\tilde{g}$ when $|y| \longrightarrow +\infty$ we have two cases:

³Historically people proved a lot of results exploiting complex analysis. The breakthrough will arrive with the Calderón-Zygmund theory, which permitted to prove the results without using complex analysis.

- $\tilde{g}(y) \rightarrow 0$, then for every $M > 0$ we define $A_M = \{z = x + iy \in \Sigma : |y| > M\}$ and we consider $F_M = F|_{\Sigma \setminus A_M}$. Since F_M is analytic on $\Sigma \setminus A_M$ both the maximum and minimum of the function must stay on $\partial(\Sigma \setminus A_M)$. Given that $g(y) \rightarrow 0$ the maximum of $|F_M|$ must stay on the vertical boundary, so

$$|F(z)| \leq 1 \quad \forall z \in \Sigma$$

- $\tilde{g}(y) \not\rightarrow 0$, then we consider the perturbed function

$$F_\epsilon(z) = F(z)e^{\epsilon(z^2-1)}$$

and we notice that:

$$|F_\epsilon(it)| = |F(it)|e^{-\epsilon(t^2+1)} \leq 1$$

$$|F_\epsilon(1+it)| = |F(1+it)|e^{\epsilon(-t^2-2it)} = |F(1+it)|e^{-\epsilon 2it}|e^{-\epsilon t^2}| \leq 1.$$

Furthermore, $\forall \epsilon > 0$

$$\tilde{g}_\epsilon(y) \rightarrow 0$$

as $|y| \rightarrow \infty$, indeed

$$\begin{aligned} \tilde{g}_\epsilon(y) &= \sup_{0 \leq x \leq 1} |F_\epsilon(x+iy)| \\ &= \sup_{0 \leq x \leq 1} |F(x+iy)|e^{\epsilon(x^2-y^2+2ixy-1)} \\ &= \sup_{0 \leq x \leq 1} |F(x+iy)|e^{\epsilon(x^2-1)}|e^{-\epsilon y^2}|e^{\epsilon(2ixy)} \\ &\leq C|e^{-\epsilon y^2}| \rightarrow 0. \end{aligned}$$

Thanks to the previous point we have proved that

$$|F_\epsilon(z)| \leq 1$$

for every $z \in \Sigma$. By letting $\epsilon \rightarrow 0$, since $F_\epsilon(z) \rightarrow F(z)$, we get the thesis.

□

Theorem 1.2.2 – Riesz-Thorin Theorem

Let $p_0 \leq p_1$, $q_0 \leq q_1$. Given a linear operator T where

$$T : L^{p_0} \rightarrow L^{q_0}$$

$$T : L^{p_1} \rightarrow L^{q_1}$$

such that there exist constants M_0 and M_1 which make valid the inequalities

$$\|Tf\|_{L^{q_0}} \leq M_0 \|f\|_{L^{p_0}}$$

$$\|Tf\|_{L^{q_1}} \leq M_1 \|f\|_{L^{p_1}}.$$

Then, for all p and q of the form

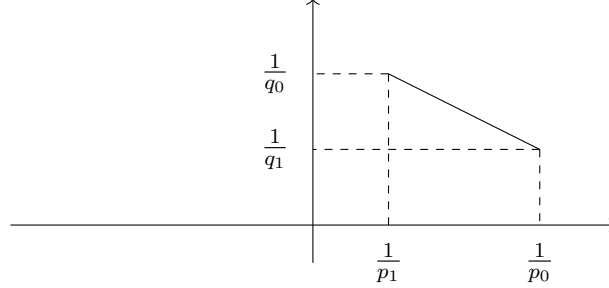
$$\begin{cases} \frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1} \\ \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1} \end{cases}$$

for $t \in (0, 1)$, we have

$$\|Tf\|_{L^q} \leq M_0^{1-t} M_1^t \|f\|_{L^p}.$$

Osservazione 1.2.1

Let us stop for a second and admire the beauty of this theorem. It states that if we have a linear continuous operator T and we have a control of the norm of $\|Tf\|$ for (p_0, q_0) and (p_1, q_1) we also have a control for every pair of exponents in the segment



Proof. By linearity the proof is completed if we show that

$$\|Tf\|_{L^{q_t}} \leq M_0^{1-t} M_1^t. \quad (1.4)$$

for $f \in L^{p_t}$ with $\|f\|_{L^{p_t}} = 1$. We start by proving the theorem for all simple function, the step from simple to a general function is made with a classical argument of density. Since $\|Tf\|_{L^q} = \sup_{\|g\|_{L^{q'}}} |\langle Tf, g \rangle|$ equation (1.4) can be written as

$$|\langle Tf, g \rangle| \leq M_0^{1-t} M_1^t$$

for f, g simple functions. Let

$$f = \sum_k \alpha_k e^{i\omega_k} \chi_{A_k}$$

$$g = \sum_k \beta_k e^{i\theta_k} \chi_{B_k}$$

where $\alpha_k, \beta_k \in \mathbb{R}_+$, $\omega_k, \theta_k \in \mathbb{R}$ and $\{A_k\}_k, \{B_k\}_k$ families of pairwise disjoint subsets of $\mathbb{R}^d$ with finite measure.

Idea Extend the definition of $|\langle Tf, g \rangle|$ to a function

$$\Sigma \ni z \rightarrow |\langle Tf, g \rangle(z)|$$

and apply Hadamard Three lines theorem. ┘

Define

$$p(z) = \frac{p}{p_0}(1-z) + \frac{p}{p_1}z \quad q'(z) = \frac{q'}{q'_0}(1-z) + \frac{q}{q_1}z$$

complex exponent and extend $|\langle Tf, g \rangle|$ to

$$F(z) := |\langle T(f(z)), g(z) \rangle|$$

where

$$f(z) = \sum_k \alpha_k^{p(z)} e^{i\omega_k} \chi_{A_k} \quad g = \sum_k \beta_k^{q'(z)} e^{i\theta_k} \chi_{B_k}.$$

It is important to notice that $\forall t \in [0, 1]$ $p(t) = q'(t) = 1$, then

$$F(t) = |\langle Tf, g \rangle|$$

and that the proof is completed if theorem (1.5.1) can be applied⁴. We start by verifying the hypothesis:

1) By linearity

$$F(z) = \sum_{k=1}^m \sum_{j=1}^n \alpha_k^{p(z)} \beta_k^{q(z)} e^{i\omega_k} e^{i\theta_k} \int_{\mathbb{R}^d} T(\chi_{A_k}) \chi_{B_k}$$

and hence F is continuous in Σ , analytic in $\overset{\circ}{\Sigma}$ and bounded, since $\alpha_k, \beta_j > 0$.

2) By direct computation

$$\begin{aligned} \|f(it)\|_{L^{p_0}}^{p_0} &= \sum |\alpha_k^{p(it)}|^{p_0} \mu(A_k) \\ &= \sum \alpha_k^{\Re(p(it))p_0} \mu(A_k) = \sum \alpha_k^p \mu(A_k) = \|f\|_{L^p}^p \\ \|g(it)\|_{L^{q'_0}}^{q'_0} &= \|g\|_{L^{q'}}^{q'} \\ \|f(1+it)\|_{L^{p_1}}^{p_1} &= \|f\|_{L^p}^p \\ \|g(1+it)\|_{L^{q'_1}}^{q'_1} &= \|g\|_{L^{q'}}^{q'} \end{aligned}$$

where we have made the calculation just for the first equality, the others are similar. We can now study the boundedness of F on the boundary of the strip as

$$\begin{aligned} |F(it)| &\leq \underbrace{\|Tf(it)\|_{L^{q_0}} \|g(it)\|_{L^{q'_0}}}_{H\ddot{o}} \\ &\leq M_0 \|f(it)\|_{L^{p_0}} \|g(it)\|_{L^{q'_0}} \leq M_0 \\ |F(1+it)| &\leq \underbrace{\|Tf(1+it)\|_{L^{q_1}} \|g(1+it)\|_{L^{q'_1}}}_{H\ddot{o}} \\ &\leq M_1 \|f(1+it)\|_{L^{p_1}} \|g(1+it)\|_{L^{q'_1}} \leq M_1. \end{aligned}$$

The hypothesis are verified and then, by taking $z \in \mathbb{R}$ the proof for simple function is completed. In order to move from simple functions to general L^p functions we do a classical approximation argument.

□

⁴Theorem 1.5.1 give us a much stronger result. Infact, it states that $F(z) \leq 1 \forall z \in \Sigma$, while we just care for $z \in \Sigma \cap \{\Im(z) = 0\}$.

Corollary 1.2.3

Considering $\mathcal{F}: L^1 \rightarrow L^\infty$ and $\mathcal{F}: L^2 \rightarrow L^2$, we can define the Fourier transform from L^p to L^q for all $1 \leq p \leq 2$ (q is the conjugate exponent of p). Furthermore,

$$\|\mathcal{F}f\|_{L^q} \leq \|f\|_{L^p}.$$

Proof. It follows directly from Riesz-Thorin theorem: we know that

$$\begin{aligned} \|\hat{f}\|_{L^\infty} &\leq \|f\|_{L^1} \\ \|\hat{f}\|_{L^2} &\leq \|f\|_{L^2} \end{aligned}$$

The estimate follows because

$$\begin{cases} \frac{1-t}{\infty} + \frac{t}{2} = \frac{1}{q} \\ \frac{1-t}{1} + \frac{t}{2} = \frac{1}{p} \end{cases} \Rightarrow \begin{cases} \frac{1}{q} = \frac{t}{2} \\ 1 - \frac{t}{2} = \frac{1}{p} \end{cases} \Rightarrow 1 = \frac{1}{p} + \frac{1}{q}$$

□

Osservazione 1.2.2

The estimate given by the corollary is called *Hausdorff-Young inequality*⁵: for $1 \leq p \leq 2$ and $\frac{1}{p} + \frac{1}{q} = 1$,

$$\|\hat{u}\|_{L^q} \leq \|u\|_{L^p}.$$

Corollary 1.2.4 – Young's inequality

Let us consider $p_1, p_2, r \in [1, +\infty]$ such that $\frac{1}{p_1} + \frac{1}{p_2} = 1 + \frac{1}{r}$. It follows that $\forall f \in L^{p_1}(\mathbb{R}^d)$ and $\forall g \in L^{p_2}(\mathbb{R}^d)$

$$\|f * g(x)\|_{L^r(\mathbb{R}^d)} \leq \|f\|_{L^{p_1}(\mathbb{R}^d)} \|g\|_{L^{p_2}(\mathbb{R}^d)}.$$

Proof. Let us consider p and its conjugate exponent p' . It is known that

$$\|f * g\|_{L^p} \leq \|f\|_{L^p} \|g\|_{L^1} \quad \text{and} \quad \|f * g\|_{L^\infty} \leq \|f\|_{L^p} \|g\|_{L^{p'}}.$$

Given $f \in L^p$ let us consider the operator $T: g \rightarrow f * g$. We have that

$$\|Tg\|_{L^\infty} \leq \underbrace{\|f\|_{L^p}}_{=M_0} \|g\|_{L^{p'}} \quad \text{and} \quad \|Tg\|_{L^1} \leq \underbrace{\|f\|_{L^p}}_{=M_1} \|g\|_{L^1}.$$

For Riesz-Thorin we have that

$$\|Tg\|_{L^R} \leq \underbrace{\|f\|_{L^p}}_{=M_0^{1-t} M_1^t} \|g\|_{L^S}$$

where

$$\begin{cases} \frac{1}{R} = \frac{1-t}{\infty} + \frac{t}{p} \\ \frac{1}{S} = \frac{1-t}{p'} + \frac{t}{1} = 1 + \frac{t-1}{p} \end{cases}.$$

⁵One can see that the optimal constant is not 1!

This means

$$\frac{1}{s} + \frac{1}{p} = 1 + \frac{1}{R}$$

hence the Young inequality is valid.

□

1.2.4 A generalisation of Riesz-Thorin theorem

We can generalise the Riesz-Thorin theorem to particular families of operators. The difficulty of this generalisation rely on a refined version of the Three Lines H. Theorem. First let us give the following definition:

Definition 1.2.5 – Admissible family of operators

Let $\Sigma = \{z \in \mathbb{C} : 0 \leq \Re(z) \leq 1\}$ be the strip in the complex plane and T_z a family of operators parameterised on Σ . We say the family $\{T_z\}$ is admissible if we have the following conditions:

- The function

$$\begin{aligned} R : \Sigma &\longrightarrow \mathbb{R} \\ z &\longmapsto \int T_z(f)gdx \end{aligned}$$

is analytic in $\mathring{\Sigma}$ and continuous in Σ , where f, g are finitely simple functions considered on two σ -finite measure space.

- There exists $a < \pi$ such that

$$\log \left(\left| \int T_z(f)gdx \right| \right) \leq C(f, g)e^{a|y|}$$

where $z = x + iy$.

Lemma 1.2.6 – Hirshman lemma

If we have a function F which is analytic on $\mathring{\Sigma}$, continuous in Σ and a constant $a < \pi$ such that

$$\sup_{\substack{0 \leq x \leq 1 \\ -\infty < y < +\infty}} e^{-a|y|} \log(|F(x + iy)|) < +\infty.$$

Then

$$^a \log |F(x)| \leq \frac{1}{2} \sin(\pi x) \int_{-\infty}^{+\infty} \frac{\log(|F(iy)|)}{\cosh(\pi y) - \cos(\pi x)} + \frac{\log(|F(1 + iy)|)}{\cosh(\pi y) + \cos(\pi x)} dy.$$

^aNotice that, on the contrary of the previous result, we have a control on F just on $\Sigma \cap \{y = 0\}$.

Osservazione 1.2.3

The condition on the growth of F translate into: there exist a constant M such that

$$|F(x + iy)| < e^{Me^{a|y|}}$$

this means that the growth must be less then double exponential.

Osservazione 1.2.4

If we suppose that both $\log(|F(iy)|)$ and $\log(|F(1 + iy)|)$ are constant or bounded we get back $|F(x)| \leq M_0^{1-x} M_1^x$, which recalls Riesz-Thorin.

Osservazione 1.2.5

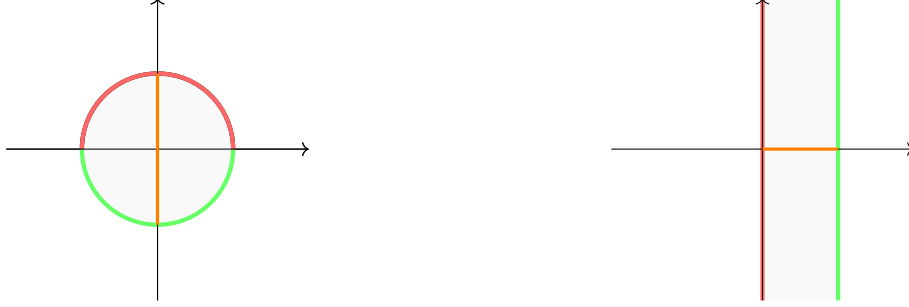
We recall that

$$\frac{1}{2} \int_{-\infty}^{+\infty} \frac{\sin(\pi x)}{\cosh(\pi y) - \cos(\pi x)} dy = 1 - x \quad \text{and} \quad \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\sin(\pi x)}{\cosh(\pi y) + \cos(\pi x)} dy = x$$

so the thesis can be seen as an interpolation of F using his value on the boundary.

Proof.

Idea The idea is to transform the strip Σ into $S = \{|z| \leq 1, z \neq \pm 1\}$ by a biholomorphism.



Once we have considered F on S we can use the Poisson Formula to control the value of the function inside the ball given the value on the boundary. $\lrcorner$

Let $h : S \rightarrow \Sigma$ be the biholomorphism defined as

$$h(z) = \frac{1}{\pi i} \log \left(i \frac{1+z}{1-z} \right)$$

and consider the function $G(\xi) := F(h^{-1}(\xi))$ defined on S . Since our goal is to get a control of F given the value on $\partial\Sigma$ we want to link G to its value on $\partial(h^{-1}\Sigma) = \partial S$. The tool we are going to use is the *Poisson's Formula*. It states: given $u : \{z \in \mathbb{C} : |z| < 1\} \rightarrow \mathbb{C}$ holomorphic then

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{R - r^2}{R^2 - 2rR \cos(\theta - \phi) + r^2} u(Re^{i\phi}) d\phi \quad (1.5)$$

for every $R < 1$. If instead we consider subharmonic function u on D that is continuous on the circle $\{|z| = R\}$ then a formula similar to (1.5) holds:

$$u(re^{i\theta}) \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{R - r^2}{R^2 - 2rR \cos(\theta - \phi) + r^2} u(Re^{i\phi}) d\phi. \quad (1.6)$$

We are particularly interested in this second version, we just need to find a subharmonic function. The choice will be $\log(|G(z)|)$ ⁶. We then have

$$\log(G(re^{i\theta})) \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{R - r^2}{R^2 - 2rR \cos(\theta - \phi) + r^2} \log(|G(Re^{i\phi})|) d\phi$$

and we want to let $R \rightarrow 1$. In order to do that we use the Lebesgue dominated convergence theorem, since we have the pointwise convergence almost everywhere of

$$P_{r,R} * \log(|G(Re^{i\theta})|) \rightarrow P_{r,1} * \log(|G(e^{i\theta})|).$$

⁶Recall that given an harmonic function u the map $\log(u)$ is subharmonic.

Furthermore, from the hypotesis, we have

$$\begin{aligned}
\frac{R - r^2}{R^2 - 2rR \cos(\theta - \phi) + r^2} \log(G(Re^{i\phi})) &\leq \log(|G(Re^{i\phi})|) \\
&= \log(|F(h(\xi))|) \\
&\leq Ae^a |\Im(h(\xi))| \\
&= Ae^a \left| \Im\left(\frac{1}{\pi i} \log\left(i \frac{1+\xi}{1-\xi}\right)\right) \right| \\
&= Ae^{\frac{a}{\pi}} \left| \log\left|\frac{1+\xi}{1-\xi}\right| \right| \\
&= \chi_{\log>0} Ae^{\frac{a}{\pi}} \left| \log\left|\frac{1+\xi}{1-\xi}\right| \right| + \chi_{\log<0} Ae^{\frac{a}{\pi}} \left| \log\left|\frac{1+\xi}{1-\xi}\right| \right| \\
&= \chi_{\log>0} A \left| \frac{1+\xi}{1-\xi} \right|^{\frac{a}{\pi}} + \chi_{\log<0} A \left| \frac{1-\xi}{1+\xi} \right|^{\frac{a}{\pi}}
\end{aligned}$$

Since $a < \pi$ for hypotesis both functions are bounded. We can now take the limit and get

$$\log(F(h(re^{i\theta}))) \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1 - r^2}{1 - 2r \cos(\theta - \phi) + r^2} \log(|F(h(e^{i\phi}))|) d\phi. \quad (1.7)$$

The main steps of the proof are completed, now it is just a matter of going back to the strip and change variable correctly.

Setting $x = h(re^{i\theta}) \in [0, 1]$, we obtain that

$$re^{i\theta} = h^{-1}(x) = \frac{e^{\pi i x} - i}{e^{\pi i x} + i} = -i \frac{\cos(\pi x)}{1 + \sin(\pi x)} = \frac{\cos(\pi x)}{1 + \sin(\pi x)} e^{-i \frac{\pi}{2}}$$

from which

$$r = \frac{\cos(\pi x)}{1 + \sin(\pi x)} \quad \theta = -\frac{\pi}{2}$$

if $x \in (0, \frac{1}{2}]$ and

$$r = -\frac{\cos(\pi x)}{1 + \sin(\pi x)} \quad \theta = \frac{\pi}{2}$$

if $x \in (\frac{1}{2}, 1)$. In either case we deduce that

$$\frac{1 - r^2}{1 - 2r \cos(\theta - \phi) + r^2} = \frac{\sin(\pi x)}{1 + \cos(\pi x) \sin(\phi)}.$$

Using this we write (1.7) as

$$\log(F(h(re^{i\theta}))) \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\sin(\pi x)}{1 + \cos(\pi x) \sin(\phi)} \log(|F(h(e^{i\phi}))|) d\phi$$

We now split the domain of integration in $[-\pi, 0) \sqcup (0, \pi]$ and change variables. On the interval $[-\pi, 0)$ we use the change of variable

$$iy = h(e^{i\phi})$$

from which

$$\sin(\phi) = \frac{-1}{\cosh(\pi y)} \quad d\phi = \frac{-\pi}{\cosh(\pi y)} dy.$$

Since as $\phi \in [-\pi, 0)$ y varies in $(-\infty, \infty)$ the first part of the integral become

$$\frac{1}{2\pi} \int_{-\pi}^0 \frac{\sin(\pi x)}{1 + \cos(\pi x) \sin(\phi)} \log(|F(h(e^{i\phi}))|) d\phi = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\sin(\pi x)}{\cosh(\pi y) - \cos(\pi x)} \log(|F(iy)|) dy.$$

On the interval $(0, \pi]$ we use the change of variable

$$1 + iy = h(e^{i\phi})$$

from which

$$\sin(\phi) = \frac{1}{\cosh(\pi y)} \quad d\phi = \frac{\pi}{\cosh(\pi y)} dy.$$

Since as $\phi \in (0, \pi]$ y varies in $(-\infty, \infty)$ the second part of the integral become

$$\frac{1}{2\pi} \int_0^\pi \frac{\sin(\pi x)}{1 + \cos(\pi x) \sin(\phi)} \log(|F(h(e^{i\phi}))|) d\phi = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\sin(\pi x)}{\cosh(\pi y) + \cos(\pi x)} \log(|F(1+iy)|) dy.$$

By putting the two part of the integral back together we conclude the proof. $\square$

Theorem 1.2.7 – Interpolation of analytic family of operators

Let $\{T_z\}_{z \in \Sigma}$ be a family of operators such that

$$\begin{cases} \|T_{iy}f\|_{L^{q_0}} \leq M_0(y)\|f\|_{L^{p_0}} \\ \|T_{1+iy}f\|_{L^{q_1}} \leq M_1(y)\|f\|_{L^{p_1}} \end{cases}$$

where there exists $b < \pi$ such that

$$\sup_{-\infty < y < +\infty} e^{-b|y|} \log(M_j(y)) < +\infty.$$

Then for all q_t, p_t of the form

$$\begin{cases} \frac{1}{p_t} = \frac{1-t}{p_0} + \frac{t}{p_1} \\ \frac{1}{q_t} = \frac{1-t}{q_0} + \frac{t}{q_1} \end{cases}$$

where $t \in (0, 1)$, we have that

$$\|T_t f\|_{L^{q_t}} \leq M(t)\|f\|_{L^{p_t}}$$

where

$$M(t) \leq \exp \left(\frac{\sin(\pi t)}{2} \int_{-\infty}^{+\infty} \frac{\log(M_0(y))}{\cosh(\pi y) - \cos(\pi t)} + \frac{\log(M_1(y))}{\cosh(\pi y) + \cos(\pi t)} dt \right).$$

Proof. The proof is identical to the one of Riesz-Thorin Theorem.

Fix $\theta \in (0, 1)$ and also fix simple functions f, g such that

$$\|f\|_{L^p} = \|g\|_{L^{q'}} = 1.$$

Let

$$f = \sum_k \alpha_k e^{i\omega_k} \chi_{A_k} \quad g = \sum_k \beta_k e^{i\theta_k} \chi_{B_k}$$

where $\alpha_k, \beta_k \in \mathbb{R}_+$, $\omega_k, \theta_k \in \mathbb{R}$ and $\{A_k\}_k, \{B_k\}_k$ families of pairwise disjoint subsets of $\mathbb{R}^d$ with finite measure. Let f_z and g_z be as the proof of theorem 1.2.3. Define

$$F(z) = \int_{\mathbb{R}^d} T_z(f_z) g_z.$$

It follows from the assumption about $\{T_z\}_z$ that $F(z)$ is an analytic function that satisfies the hypothesis of 1.2.4. Moreover,

$$\begin{aligned} \|f_{iy}\|_{L^{p_0}}^{p_0} &= \|f_{iy}\|_{L^p}^p = 1 = \|g_{iy}\|_{L^{q'}}^{q'} = \|f_{iy}\|_{L^{q'_0}}^{q'_0} \\ \|f_{1+iy}\|_{L^{p_0}}^{p_0} &= \|f_{1+iy}\|_{L^p}^p = 1 = \|g_{1+iy}\|_{L^{q'}}^{q'} = \|f_{1+iy}\|_{L^{q'_0}}^{q'_0} \end{aligned}$$

when $y \in \mathbb{R}$. Now thanks to this relation, Holder Inequality and the hypothesis we get

$$\begin{aligned} |F(iy)| &\leq \|T_{iy}(f_{iy})\|_{L^{q_0}} \|g_{iy}\|_{L^{q'_0}} \\ &\leq M_0(y) \|f_{iy}\|_{L^{p_0}} \|g_{iy}\|_{L^{q'_0}} = M_0(y) \end{aligned}$$

for all y real. Similarly

$$\begin{aligned} |F(1+iy)| &\leq \|T_{1+iy}(f_{1+iy})\|_{L^{q_1}} \|g_{1+iy}\|_{L^{q'_1}} \\ &\leq M_1(y) \|f_{1+iy}\|_{L^{p_1}} \|g_{1+iy}\|_{L^{q'_1}} = M_1(y) \end{aligned}$$

for all y real. Therefor the hypothesis of Lemma (H) are satisfied. We conclude that

$$\left| \int_{\mathbb{R}^d} T_\theta(f) g \right| = |F(\theta)| \leq M(\theta)$$

where $M(x)$ is the function given in the hypothesis of the theorem.

We conclude by taking the supremum over all simple function g with $L^{q'}$ norm equal to one. $\square$

Corollary 1.2.8 – Stein Theorem

Take^a K_0, K_1, u_0, u_1 measurable functions. Suppose that

$$\begin{cases} \|K_0 T(f)\|_{L^{q_0}} \leq M_0 \|f u_0\|_{L^{p_0}} \\ \|K_1 T(f)\|_{L^{q_1}} \leq M_1 \|f u_1\|_{L^{p_1}} \end{cases}$$

Then for all q_t, p_t of the form

$$\begin{cases} \frac{1}{p_t} = \frac{1-t}{p_0} + \frac{t}{p_1} \\ \frac{1}{q_t} = \frac{1-t}{q_0} + \frac{t}{q_1} \end{cases}$$

where $t \in (0, 1)$, it holds

$$\|K_t T(f)\|_{L^{q_t}} \leq M_0^{1-t} M_1^t \|u_t f\|_{L^{p_t}}$$

where $K_t = K_0^{1-t} K_1^t$ and $u_t = u_0^{1-t} u_1^t$.

^aWith this theorem we can have an interpolation of operators with weighted condition.

1.3 Marcinkiewitz theorem

Until now we have successfully proved two theorems in Interpolation Theory using tool of Complex Analysis. In this section we will proceed to prove a similar result only using Real Analysis results.

Definition 1.3.1 – Distribution function

Given a measurable function f we define the *distribution function* of f as

$$d_f(\alpha) = \mu\{x \in \mathbb{R}^d : |f(x)| > \alpha\}.$$

The distribution function d_f provides information about the size of f but not about the behaviour of f itself near any given point. It follows from definition that d_f is a non increasing function of α .

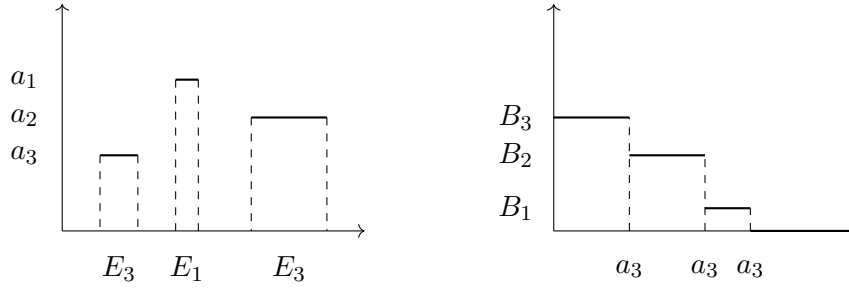


Figure 1.1: The graph of a simple function $f(x) = \sum_{j=1}^3 a_j \chi_{E_j}$ and its distribution function $d_f(\alpha)$. Here $B_j = \sum_{k=1}^j \mu(E_k)$.

The distribution function can be used to rewrite the L^p -norm of a function f as follow.

Proposition 1.3.2 – Layer-Cake decomposition

For every $p \in (0, +\infty)$ it holds that

$$\|f\|_{L^p}^p = p \int_0^{+\infty} \alpha^{p-1} d_f(\alpha) d\alpha.$$

Proof. By applying Fubini we get

$$\begin{aligned} p \int_0^{+\infty} \alpha^{p-1} d_f(\alpha) d\alpha &= p \int_0^{+\infty} \alpha^{p-1} \mu\{|f| > \alpha\} d\alpha \\ &= p \int_0^{+\infty} \alpha^{p-1} \left(\int_{\mathbb{R}^d} \chi_{\{|f| > \alpha\}}(x) dx \right) d\alpha \\ &= \int_{\mathbb{R}^d} \left(\int_0^{|f|} p \alpha^{p-1} d\alpha \right) dx \\ &= \int_{\mathbb{R}^d} |f|^p dx. \end{aligned}$$

□

Definition 1.3.3 – Weak L^p spaces

For all $p \in (0, +\infty)$ we define the *weak L^p* space, also denoted by $L^{p,\infty}$, as the set of measurable function such that

$$\|f\|_{L^{p,\infty}} = \inf \left\{ c : d_f(\alpha) \leq \frac{c^p}{\alpha^p} \quad \forall \alpha > 0 \right\} < +\infty.$$

We have that $L^{p,\infty}$ are quasi-normed spaces:

$$\|f + g\|_{L^{p,\infty}} \leq C_p(\|f\|_{L^{p,\infty}} + \|g\|_{L^{p,\infty}})$$

where $C_p = \max(2, 2^{\frac{1}{p}})$.

Osservazione 1.3.1

It is possible to generalize this concept by defining the spaces $L^{p,q}$. They are called Lorentz Spaces.

Osservazione 1.3.2

The space weak $L^{\infty,\infty}$ is by definition L^∞ .

Proposition 1.3.4

For all $p \in (0, +\infty)$

$$\|f\|_{L^{p,\infty}} \leq \|f\|_{L^p}$$

hence $L^p \subset L^{p,\infty}$. Moreover $\forall p \exists f \in L^{p,\infty}$ such that $f \notin L^p$.

Proof. By the Layer-Cake decomposition, for every $\alpha_0 > 0$, holds that

$$\begin{aligned} \|f\|_{L^p}^p &= p \int_0^{+\infty} \alpha^{p-1} d_f(\alpha) d\alpha \\ &\geq p \int_0^{\alpha_0} \alpha^{p-1} d_f(\alpha) d\alpha \\ &\geq p \int_0^{\alpha_0} \alpha^{p-1} d_f(\alpha_0) d\alpha = \mu\{|f| > \alpha_0\} \alpha_0^p \end{aligned}$$

Rearranging the terms we get the *Chebychev inequality*

$$\mu\{|f| > \alpha_0\} \leq \frac{\|f\|_{L^p}^p}{\alpha_0^p}.$$

Since $\|f\|_{L^p}^p < \infty$

$$\|f\|_{L^{p,\infty}} \leq \|f\|_{L^p}.$$

To show that $L^{p,\infty} \neq L^p$ we might choose $f = |x|^{-\frac{d}{p}}$. This function clearly does not stay in L^p , yet

$$\mu \left\{ |x|^{-\frac{d}{p}} > \alpha \right\} = \mu \left(B \left(0, \frac{1}{\alpha^{\frac{p}{d}}} \right) \right) = \frac{\omega_d}{\alpha^p}$$

where ω_d is the volume of the unit ball. □

Corollary 1.3.5 – Chebychev Inequality

We also have the Chebychev inequality:

$$\mu\{|f| > \lambda\} \leq \frac{\|f\|_{L^p}^p}{\lambda^p}.$$

Definition 1.3.6

Let us consider an operator T . We say that T is

- *Sublinear*: if for all f, g

$$|T(f+g)| \leq |T(f)| + |T(g)|$$

- *Strong (p, p) -continuous*: if

$$\|Tf\|_{L^p} \leq c\|f\|_{L^p}.$$

- *Weak (p, p) -continuous*: if

$$\mu\{|Tf| > \lambda\} \leq c \frac{\|f\|_{L^p}^p}{\lambda^p}.$$

Osservazione 1.3.3

It is clear that if T is strong (p, p) continuous, then it is also weak (p, p) : indeed by Chebychev inequality we have

$$\mu\{|Tf| > \lambda\} \leq \frac{\|Tf\|_{L^p}^p}{\lambda^p} \leq \frac{c^p \|f\|_{L^p}^p}{\lambda^p}.$$

Theorem 1.3.7 – Marcinkiewitz theorem

Let us consider p_0, p_1 such that $1 \leq p_0 < p_1 < +\infty$. If $T : L^{p_0} + L^{p_1} \rightarrow \{\text{measurable functions}\}$ is sublinear, (p_0, p_0) -weak and (p_1, p_1) -weak continuous, then T is (p, p) -strong continuous for all $p \in (p_0, p_1)$.

Osservazione 1.3.4

The theorem is still true taking $p_1 = +\infty$ and asking T to be (p_1, p_1) -strong continuous.

Proof. Given $f \in L^p$ with $p \in (p_0, p_1)$ we want to show that exists a constant $A < \infty$ such that

$$\|T(f)\|_{L^p} \leq A\|f\|_{L^p}.$$

Since T is (p_0, p_0) -weak and (p_1, p_1) -weak continuous there exists two positive constants A_0, A_1 such that $\|T(f)\|_{L^{p_0, \infty}} \leq A_0\|f\|_{L^{p_0}}$ and $\|T(f)\|_{L^{p_1, \infty}} \leq A_1\|f\|_{L^{p_1}}$.

Fix $\alpha, \lambda > 0$ and we split f in two functions

$$h = \begin{cases} f & |f| > \frac{\lambda}{\alpha} \\ 0 & \text{otherwise} \end{cases} \quad g = \begin{cases} f & |f| \leq \frac{\lambda}{\alpha} \\ 0 & \text{otherwise} \end{cases}$$

such that $h \in L^{p_0}$ and $g \in L^{p_1}$. Infact h , the unbounded part of f has finite support

measure (otherwise $f \notin L_0^p$), then

$$\int_{\mathbb{R}^d} h^{p_0} dx = \int_{\mathbb{R}^d} \chi_{\{supp(h)\}} h^{p_0} \leq \left(\int_{\mathbb{R}^d} |f|^p \right)^{\frac{p_0}{p}} \mu(\{supp(h)\})^{\frac{p-p_0}{p}}$$

where the last inequality is Holder with $\frac{p}{p_0} \cdot \frac{p}{p-p_0}$. For the bounded part of f, g , we get

$$\int_{\mathbb{R}^d} |g|^{p_1} = \int_{\mathbb{R}^d} |g|^{p_1-p} |g|^p \leq \int_{\mathbb{R}^d} \left(\frac{\lambda}{\alpha} \right)^{p_1-p} |f|^p = \left(\frac{\lambda}{\alpha} \right)^{(p_1-p)} \|f\|_{L^p}^p$$

which is finite.

By the sublinearity property of T we obtain that

$$|T(f)| \leq |T(h)| + |T(g)|$$

which implies

$$\{x : |T(f)(x)| > \lambda\} \subseteq \left\{x : |T(h)(x)| > \frac{\lambda}{2}\right\} \cup \left\{x : |T(g)(x)| > \frac{\lambda}{2}\right\}$$

and therefore

$$\|T(f)\|_{L^p}^p \leq p \int_0^\infty \lambda^{p-1} \mu \left\{ |T(h)| > \frac{\lambda}{2} \right\} d\lambda + p \int_0^\infty \lambda^{p-1} \mu \left\{ |T(g)| > \frac{\lambda}{2} \right\} d\lambda$$

By direct computation we get

$$\begin{aligned} p \int_0^\infty \lambda^{p-1} \mu \left\{ |T(h)| > \frac{\lambda}{2} \right\} d\lambda &\leq p A_0^{p_0} 2^{p_0} \int_0^\infty \lambda^{p-1} \frac{\|h\|_{L^{p_0}}^{p_0}}{\lambda^{p_0}} d\lambda \\ &\leq p A_0^{p_0} 2^{p_0} \int_0^\infty \lambda^{p-p_0-1} \int_{\{|f|>\frac{\lambda}{\alpha}\}} |f|^{p_0} dx d\lambda \\ &= p A_0^{p_0} 2^{p_0} \int_{\mathbb{R}^d} |f|^{p_0} \int_0^{\alpha|f|} \lambda^{p-p_0-1} d\lambda dx \\ &= \frac{p}{p-p_0} A_0^{p_0} 2^{p_0} \alpha^{p-p_0} \int_{\mathbb{R}^d} |f|^p dx \end{aligned}$$

and

$$\begin{aligned} p \int_0^\infty \lambda^{p-1} \mu \left\{ |T(g)| > \frac{\lambda}{2} \right\} d\lambda &\leq p A_1^{p_1} 2^{p_1} \int_0^\infty \lambda^{p-1} \frac{\|g\|_{L^{p_1}}^{p_1}}{\lambda^{p_1}} d\lambda \\ &\leq p A_1^{p_1} 2^{p_1} \int_0^\infty \lambda^{p-p_1-1} \int_{\{|f|<\frac{\lambda}{\alpha}\}} |f|^{p_1} dx d\lambda \\ &= p A_1^{p_1} 2^{p_1} \int_{\mathbb{R}^d} |f|^{p_1} \int_0^{\alpha|f|} \lambda^{p-p_1-1} d\lambda dx \\ &= \frac{p}{p_1-p} A_1^{p_1} 2^{p_1} \alpha^{p-p_1} \int_{\mathbb{R}^d} |f|^p dx \end{aligned}$$

Putting back together the integrals we obtain

$$\|T(f)\|_{L^p}^p \leq \left(\frac{p}{p-p_0} A_0^{p_0} 2^{p_0} \alpha^{p-p_0} + \frac{p}{p-p_1} A_1^{p_1} 2^{p_1} \alpha^{p-p_1} \right) \|f\|_{L^p}^p.$$

In order to get the sharpest constant we search for the minimum over α . This can be found by setting

$$\frac{p}{p-p_0} A_0^{p_0} 2^{p_0} \alpha^{p-p_0} = \frac{p}{p-p_1} A_1^{p_1} 2^{p_1} \alpha^{p-p_1}$$

which get us to

$$\|T(f)\|_{L^p} \leq A \|f\|_{L^p}^p$$

with

$$A = 2 \left(\frac{p}{p-p_0} + \frac{p}{p-p_1} \right)^{\frac{1}{p}} A_0^{\frac{\frac{1}{p}-\frac{1}{p_1}}{\frac{1}{p_0}-\frac{1}{p_1}}} A_1^{\frac{\frac{1}{p_0}-\frac{1}{p}}{\frac{1}{p_0}-\frac{1}{p_1}}}.$$

□

Chapter 2

The Hardy-Littlewood-Sobolev inequality

In this chapter we want to achieve some inequality of the type¹: $\left\| \frac{1}{|x|^\alpha} * f \right\|_{L^q} \preccurlyeq \|f\|_{L^p}$.

Proposition 2.0.1

Given $f \in C_c^\infty$ and $C_\alpha = \pi^{-\frac{\alpha}{2}} \Gamma\left(\frac{\alpha}{2}\right)$. If $0 \leq \alpha < d$ we have that

$$(C_\alpha |\xi|^{-\alpha} \hat{f})^\vee = C_{d-\alpha} \int_{\mathbb{R}^d} \frac{f(y)}{|x-y|^{d-\alpha}} dy.$$

Proof. Let's remember the definition of the *Gamma function*: $\Gamma(x) = \int_0^{+\infty} e^{-t} t^{x-1} dt$, which has the property that $\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin(\pi x)}$. We notice that if $0 < \alpha < d$ then

$$\begin{aligned} C_\alpha |\xi|^{-\alpha} &= \int_0^{+\infty} e^{-\pi|\xi|^2 \lambda} \lambda^{\frac{\alpha}{2}-1} d\lambda = \int_0^\infty e^{-t} \left[\frac{t}{\pi|\xi|^2} \right]^{\frac{\alpha}{2}-1} \frac{1}{\pi|\xi|^2} dt \\ &= \int_0^\infty e^{-t} t^{\frac{\alpha}{2}-1} \pi^{\frac{\alpha}{2}} |\xi|^{-\alpha} \\ &= \Gamma\left(\frac{\alpha}{2}\right) \pi^{\frac{\alpha}{2}} |\xi|^{-\alpha}. \end{aligned}$$

Thanks to this equality we have that

$$\begin{aligned} (C_\alpha |\xi|^{-\alpha} \hat{f})^\vee &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \int_0^\infty e^{-\pi|\xi|^2 \lambda} \lambda^{\frac{\alpha}{2}-1} d\lambda f(y) e^{-2\pi i \xi \cdot y} dy e^{2\pi i \xi \cdot x} d\xi \\ &= \int_0^\infty \int_{\mathbb{R}^d} \lambda^{\frac{\alpha}{2}-1} f(y) e^{-\frac{\pi(x-y)^2}{\lambda}} \lambda^{-\frac{d}{2}} d\lambda dy \\ &= \int_{\mathbb{R}^d} f(y) \int_0^\infty \varepsilon^{1-\frac{\alpha}{2}} e^{-\pi(x-y)^2 \varepsilon} \varepsilon^{\frac{d}{2}} \varepsilon^{-2} d\varepsilon dy \\ &= \int_{\mathbb{R}^d} f(y) \int_0^\infty \varepsilon^{\frac{d-\alpha}{2}-1} e^{-\pi(x-y)^2 \varepsilon} d\varepsilon dy \\ &= \int_{\mathbb{R}^d} C_{d-\alpha} |x-y|^{-(d-\alpha)} dy \\ &= C_{d-\alpha} \int_{\mathbb{R}^d} \frac{f(y)}{|x-y|^{d-\alpha}} dy. \end{aligned}$$

¹by $\preccurlyeq$ we mean that there is the inequality up to a constant, i.e. $\preccurlyeq \equiv \leq C \cdot$.

□

Osservazione 2.0.1

Since $f \in C_C^\infty$ we have that $\hat{f}$ is well defined, analytic and, for $|\xi| \rightarrow \infty$, all of its derivatives decay faster than the inverse of any polynomial² in ξ , hence $|\xi|^{-\alpha} \hat{f} \in L^1$. Since, a priori, $\int_{\mathbb{R}^d} \frac{f(y)}{|x-y|^{d-\alpha}} dy$ decays only as $|x|^{\alpha-d}$ it is not in L^p for any $p \leq 2$ but, if $0 < \alpha < \frac{d}{2}$, thanks to the Hardy-Littlewood-Sobolev inequality it is a L^2 function, hence it admits Fourier transform. This gives us the relation:

$$C_\alpha |\xi|^{-\alpha} \hat{f} = C_{d-\alpha} \left(\int_{\mathbb{R}^d} \frac{\widehat{f(y)}}{|x-y|^{d-\alpha}} dy \right).$$

Idea We can study the Poisson equation: $-\Delta u = f$ on $\mathbb{R}^3$ where $f \in C_C^\infty$. We remember that³

$$\begin{aligned} \widehat{\frac{\partial u}{\partial x_i}} &= \int \frac{\partial u}{\partial x_i}(x) e^{-2\pi i \xi \cdot x} dx = -2\pi i \xi_i \int u(x) e^{-2\pi i \xi \cdot x} \\ \left(\widehat{\frac{\partial^2 u}{\partial x_i^2}} \right) &= 4\pi^2 (\xi_i)^2 \int u(x) e^{-2\pi i \xi \cdot x}. \end{aligned}$$

So if we apply the Fourier transform to the Poisson equation we get:

$$-\widehat{\Delta u} = \hat{f} \Rightarrow 4\pi^2 |\xi|^2 \hat{u} = \hat{f} \Rightarrow \hat{u} = \frac{1}{4\pi^2} \frac{1}{|\xi|^2} \hat{f}.$$

Thanks to the previous formula, with $\alpha = 2$ and $d = 3$ we have:

$$u = \frac{1}{4\pi^2} \left(\frac{1}{|\xi|^2} \hat{f} \right)^\vee = \frac{1}{4\pi} \int \frac{f(y)}{|x-y|} dy$$

indeed:

$$\begin{aligned} \left(\frac{1}{|\xi|^2} \hat{f} \right)^\vee &= \frac{C_1}{C_2} \int_{\mathbb{R}^3} \frac{f(y)}{|x-y|} dy \\ &= \sqrt{\pi} \frac{\Gamma(\frac{1}{2})}{\Gamma(1)} \int_{\mathbb{R}^3} \frac{f(y)}{|x-y|} dy \\ &= \sqrt{\pi} \Gamma\left(\frac{1}{2}\right) \int_{\mathbb{R}^3} \frac{f(y)}{|x-y|} dy \\ &= \sqrt{\pi} \sqrt{\frac{\pi}{\sin(\frac{\pi}{2})}} \int_{\mathbb{R}^3} \frac{f(y)}{|x-y|} dy \\ &= \pi \int_{\mathbb{R}^3} \frac{f(y)}{|x-y|} dy. \end{aligned}$$

┘

²This type of function will be really useful.

³you can think that u has compact support.

2.1 The maximal function

Definition 2.1.1 – Centered maximal function

Given $f \in L^1_{loc}(\mathbb{R}^d)$ we define the *centered maximal function* Mf as

$$\begin{aligned} Mf(x) &= \sup_{\delta > 0} \text{Avg}_{B(x, \delta)} |f| \\ &= \sup_{\delta > 0} \frac{1}{\omega_d \delta^d} \int_{|y| < \delta} |f(x - y)| dy. \end{aligned}$$

The centered maximal function can be thought as the worst average of a function.

Example 2.1.2. Let us compute Mf where $f = \chi_{[a, b]}(x)$. We start by noticing that, if $x > b$, then having for $\delta > 0$ $I = (x - \delta, x + \delta)$ it holds that

$$\frac{1}{2\delta} \int_I |f(y)| dy = \frac{\delta - (x - b)}{2\delta}.$$

If we consider $\delta = x - a$ then

$$Mf(x) = \begin{cases} \frac{b-a}{2|x-b|} & x \leq a \\ 1 & a < x < b \\ \frac{b-a}{2|x-a|} & x \geq b \end{cases}.$$

Since $Mf \sim \frac{1}{|x|}$ then $Mf \notin L^1$.

As we can see, even a simple function as $\chi_{[a, b]}$ has maximal function which is not in L^1 . The reality is even worse.

Proposition 2.1.3

Given $f \in L^1_{loc}(\mathbb{R}^d)$ if $Mf \in L^1(\mathbb{R}^d)$ then $f \equiv 0$.

Proof. It is clear that $B(0, R) \subseteq B(x, \|x\| + R)$ for all $x \in \mathbb{R}^d$ and $R \in \mathbb{R}^+$. So for all $R \in \mathbb{R}^+$ it holds:

$$\begin{aligned} Mf(x) &\geq \frac{1}{\mu(B(x, \|x\| + R))} \int_{B(x, \|x\| + R)} |f(y)| dy \\ &\geq \frac{1}{\omega_d (\|x\| + R)^d} \int_{B(0, R)} |f(y)| dy. \end{aligned}$$

This means that if $Mf \in L^1$ then $\int_{B(0, R)} |f(y)| dy = 0$ which is $f = 0$ a.e. □

Osservazione 2.1.1

By changing slightly last proof it can be shown that if $Mf(x_0) = 0$ for some x_0 then $f \equiv 0$ a.e.

Osservazione 2.1.2

It holds that $\{Mf > \lambda\} \subseteq \mathbb{R}^d$ is an open subset, i.e. Mf is lower semi continuous.

Definition 2.1.4 – Non centered maximal function

We have

$$\begin{aligned}\tilde{M}_f(x) &= \sup_{\substack{\delta > 0 \\ |y-x| < \delta}} \text{Avg}_{B(y,\delta)} |f| \\ &= \sup_{\substack{\delta > 0 \\ |y-x| < \delta}} \frac{1}{\mu(B(y,\delta))} \int_{B(y,\delta)} |f(z)| dz.\end{aligned}$$

Example 2.1.5. The non-centered maximal function for $f = \chi_{[a,b]}$ can be obtained similarly as we have done before. After some computation we get

$$Mf(x) = \begin{cases} \frac{b-a}{|x-b|} & x \leq a \\ 1 & a < x < b \\ \frac{b-a}{|x-a|} & x \geq b \end{cases}.$$

Observe that Mf does not have a jump at $x = a$ and $x = b$.

Osservazione 2.1.3

It holds that $\{\tilde{M}_f > \lambda\} \subseteq \mathbb{R}^d$ is an open subset, i.e. $\tilde{M}_f$ is lower semi continuous.

Proof. If we consider $x \in \{\tilde{M}_f > \lambda\}$, there exists y such that

$$\frac{1}{\mu(B(y,R))} \int_{B(y,R)} |f(z)| dz > \lambda \quad \forall x \in B(y,R).$$

This means that $B(y,R) \subseteq \{\tilde{M}_f > \lambda\}$, hence the thesis. $\square$

Proposition 2.1.6

Given f we have

$$Mf \leq \tilde{M}_f \leq 2^d Mf.$$

Proof. It is clear that $Mf \leq \tilde{M}_f$. Let see $\tilde{M}_f \leq 2^d Mf$. We start by seeing that, if $x \in B(y,R)$ then $B(y,R) \subseteq B(x,2R)$. It follows that:

$$\begin{aligned}\frac{1}{\mu(B(y,R))} \int_{B(y,R)} |f(z)| dz &\leq \frac{1}{\mu(B(x,R))} \int_{B(x,2R)} |f(z)| dz \\ &= \frac{2^d}{\mu(B(x,2R))} \int_{B(x,2R)} |f(z)| dz \leq 2^d Mf.\end{aligned}$$

$\square$

Corollary 2.1.7

We have that

$$\mu\{Mf < \lambda\} \leq c \frac{\|f\|_{L^1}}{\lambda} \iff \mu\{\tilde{M}_f < \lambda\} \leq \tilde{c} \frac{\|f\|_{L^1}}{\lambda}.$$

Lemma 2.1.8 – Vitali covering lemma

Let $\{B_1, \dots, B_k\}$ be a finite collection of balls in $\mathbb{R}^d$. Then there exists a subset $\{\tilde{B}_{j_1}, \dots, \tilde{B}_{j_l}\}$ of pointwise disjoint balls such that

$$\bigcup_{i=1}^k B_i \subseteq \bigcup_{r=1}^l 3\tilde{B}_{j_r}.$$

Furthermore

$$\mu\left(\bigcup_{r=1}^l \tilde{B}_{j_r}\right) = \sum_{r=1}^l \mu(\tilde{B}_{j_r}) \geq \frac{1}{3^d} \mu\left(\bigcup_{i=1}^k B_i\right).$$

Proof. Omitted. □

Theorem 2.1.9 – Hardy-Littlewood

It holds that $1 \leq p \leq +\infty$

$$\|Mf\|_{L^p} \leq C_p \|f\|_{L^p}.$$

Proof. The thesis will follow thanks to Marcinkiewicz theorem. It is clear that $\|Mf\|_{L^\infty} \leq \|f\|_{L^\infty}$. We now want a $(1,1)$ -weak estimate. Thanks to corollary 2.1 we will show an estimate for $\tilde{M}_f$ instead that for Mf . Let us call $E_\lambda = \{\tilde{M}f > \lambda\}$. The $(1,1)$ -weak continuity will follow by an estimate of $\mu(E_\lambda)$. Since it is a measurable set (it is open) we can find $K \subseteq E_\lambda$ compact such that $\mu(K) + \varepsilon = \mu(E_\lambda)$ and $\forall x \in K \exists B_x$ such that

$$\frac{1}{\mu(B_x)} \int_{B_x} |f(z)| dz > \lambda.$$

Since K is compact we can find B_{x_i} such that $K = \bigcup_{i=1}^n B_{x_i}$. For the Vitali covering lemma we also have

$$K \subseteq \bigcup B_i \subseteq 3\tilde{B}_j$$

where $\tilde{B}_j$ are disjoint. It follows that:

$$\begin{aligned} \mu(K) &\leq \mu\left(\bigcup_i B_i\right) \leq 3^d \sum_j \mu(\tilde{B}_j) \\ &\leq \sum_j 3^d \frac{1}{\lambda} \int_{\tilde{B}_j} |f(y)| dy \\ &\leq \frac{3^d}{\lambda} \int_{\bigcup_j \tilde{B}_j} |f(y)| dy \\ &\leq \frac{3^d}{\lambda} \|f\|_{L^1(\mathbb{R}^d)}. \end{aligned}$$

By letting ε go to zero the proof is complete. □

Definition 2.1.10 – Maximal operator

If we have a family of linear operators $\{T_n\}$ we define the *maximal operator* associated to the family as

$$T^*f(x) := \sup_n |T_n f(x)|.$$

Lemma 2.1.11

Let $\{T_n\}$ be a family of linear operators, such that for each n

$$T_n : L^p \rightarrow \{\text{measurable functions}\}$$

and T^* is of weak type (p, p) . Then the following set is closed:

$$C = \{f \in L^p \text{ such that } \lim_{n \rightarrow +\infty} T_n f = f \text{ almost every } x\}.$$

Proof. Since L^p is a metric space we have that

$$C \text{ closed} \iff C \text{ sequentially closed.}$$

We want to prove that if $f_k \in C$ and $f_k \rightarrow f$ then $f \in C$. Given $\varepsilon > 0$ we have $\|f_k - f\| \leq \varepsilon$ for large enough k . Fix $\lambda > 0$ we want to prove that there exists $c \in \mathbb{R}^+$ such that:

$$\mu \left\{ x : \limsup_{n \rightarrow +\infty} |T_n f - f| > \lambda \right\} \leq c\varepsilon^p.$$

Idea The following argument is really useful and many proofs use this basic idea: if we have that $x + y + z > \lambda$ then, worst case scenario, we surely have $\begin{cases} x > \frac{\lambda}{3} \\ y > \frac{\lambda}{3} \\ z > \frac{\lambda}{3} \end{cases}$. $\lrcorner$

Thanks to the linearity of T we can write:

$$\begin{aligned} \mu \left\{ x : \limsup_{n \rightarrow +\infty} |T_n f - f| > \lambda \right\} &= \mu \left\{ x : \limsup_{n \rightarrow +\infty} |T_n f - T_n f_k + T_n f_k - f_k + f_k - f| > \lambda \right\} \\ &\leq \mu \left\{ x : \limsup_{n \rightarrow +\infty} |T_n(f - f_k)| > \frac{\lambda}{3} \right\} + \\ &\quad + \mu \left\{ x : \limsup_{n \rightarrow +\infty} |T_n f_k - f_k| > \frac{\lambda}{3} \right\} + \\ &\quad + \mu \left\{ x : \limsup_{n \rightarrow +\infty} |f_k - f| > \frac{\lambda}{3} \right\} \\ &= A + B + C \preccurlyeq \frac{\varepsilon^p}{\lambda^p} \end{aligned}$$

where the last estimate follows because:

$$\begin{aligned}
A &\leq \mu \left\{ x : |T^*(f - f_k)| > \frac{\lambda}{3} \right\} \\
&\leq C \frac{\|f - f_k\|_{L^p}^p}{\lambda^p} \\
&\leq k \frac{\varepsilon^p}{\lambda^p} \text{ thanks to the } (p, p)\text{-weak continuity} \\
B &= 0 \text{ having } f_k \in C \\
C &\leq k \frac{\varepsilon^p}{\lambda^p} \text{ thanks to the Chebychev inequality}
\end{aligned}$$

□

Corollary 2.1.12 – Lebesgue differentiation theorem

If $f \in L^1$ then

$$\lim_{r \rightarrow 0} \frac{1}{\mu(B(x, r))} \int_{B(x, r)} f(y) dy = f(x)$$

for almost every x .

Proof. Let us define the family of operators T_r where

$$T_r f(x) = \int_{B(x, r)} f(y) dy$$

and consider the maximal operator $T^* f(x)$. We notice that T^* is $(1, 1)$ -weak, indeed

$$\left| \int_{B(x, r)} f(y) dy \right| \leq \int_{B(x, r)} |f(y)| dy$$

which means that $T^* f(x) \leq M f(x)$. Having that $M f$ is $(1, 1)$ -weak, also $T^* f$ is. For the previous lemma the following is a closed set of L^1 :

$$\{f \in L^1 : \lim_{r \rightarrow 0} T_r f = f \text{ a.e.}\}.$$

We conclude because for $f \in C_C^\infty$ it is clear that $\lim_{r \rightarrow 0} T_r f = f$ and they are a dense subset of L^1 . □

Proposition 2.1.13

Given $K \in L^1$ such that $K(x) = K(\|x\|)$ is a radial, non-increasing function we have that

$$|K * f(x)| \leq \|K\|_{L^1} M f(x).$$

Proof. Let us suppose $K \in C_C^\infty$, then⁴:

$$\begin{aligned}
|K * f| &= \left| \int_{\mathbb{R}^d} K(y) f(x-y) dy \right| \\
&= \left| \int_0^{+\infty} K(r) \left(\int_{\Sigma_r} f(x-y) d\sigma_r \right) dr \right| \\
&\stackrel{PP}{=} \left| - \int_0^{+\infty} K'(r) \left(\underbrace{\int_0^r \int_{\Sigma_s} f(x-y) d\sigma_s ds}_{\int_{B(x,r)} f(y) dy} \right) dr \right| \\
&= \left| - \int_0^{+\infty} K'(r) \frac{\mu(B(x,r))}{\mu(B(x,r))} \int_{B(x,r)} f(y) dy dr \right| \\
&\leq \left| \int_0^{+\infty} -K'(r) \mu(B(x,r)) Mf(x) dr \right| \\
&\leq Mf \left| \int_0^{+\infty} -K'(r) \mu(B(x,r)) dr \right| \\
&= Mf \left| \int_0^{+\infty} -K'(r) \int_0^r \left(\int_{\Sigma_s} 1 d\sigma_s \right) ds dr \right| \\
&\stackrel{PP}{\leq} Mf \int_0^\infty |K(r)| \int_{\Sigma_r} d\sigma_r dr \\
&= Mf \|K\|_{L^1}
\end{aligned}$$

where we have used the integration by parts theorem twice. $\square$

Corollary 2.1.14

If we consider $K_\varepsilon(x) = \frac{1}{\varepsilon^d} K\left(\frac{x}{\varepsilon}\right)$ with $K \in L^1$, non increasing and radial. Then K^* is $(1, 1)$ -weak.

Proof. We have $\forall \varepsilon$

$$|K_\varepsilon * f(x)| \leq \|K_\varepsilon\|_{L^1} Mf \leq \|K\|_{L^1} Mf.$$

This implies

$$K^* f = \sup_{\varepsilon} |K_\varepsilon f| \leq \|K\|_{L^1} Mf.$$

$\square$

Corollary 2.1.15

Assume $\int K dx = 1$ then $\forall f \in L^p$ with $1 \leq p < +\infty$ we have

$$K_\varepsilon * f \xrightarrow{\varepsilon \rightarrow 0} f \text{ a.e.}$$

⁴It is important to have $-K'(r) > 0$ so that we can consider Mf without any problem.

Proof. Let us consider the maximal operator $K^*f = \sup_{\varepsilon>0} |K_\varepsilon * f|$. Let us see that it is (p, p) -weak:

$$\mu\{K^*f > \lambda\} \leq \mu\{\|K\|_{L^1} Mf > \lambda\} \leq \frac{c\|f\|_{L^p}^p \|K\|_{L^1}^p}{\lambda^p}.$$

We conclude thanks to the lemma above. $\square$

Osservazione 2.1.4

The previous result still holds, in a certain way, even if K which is not radial nor non-increasing: it is sufficient to consider a *radially symmetric majorant*, i.e. a function K_0 such that K_0 is radially symmetric, non increasing and

$$|K(x)| \leq |K_0(x)|.$$

Using the result on K_0 we get

$$|K * f| \leq \|K_0\|_{L^1} Mf(x).$$

2.1.1 Applications

The Lemma (2.1) has some non-obvious consequences in the theory of PDEs. Let us study the case of the *heat equation* and *Schrödinger's equation*.

Definition 2.1.16 – Heat Kernel

Given the *heat equation* with initial datum $f \in L^2(I)$, i.e.:

$$\begin{cases} \partial_t u = \Delta u \\ u(x) = f \end{cases}$$

we say u is a solution if

$$u = H_t * f,$$

where we define the *heat kernel* as

$$H_t = \frac{1}{(4\pi t)^{\frac{d}{2}}} e^{-\frac{\pi|x|^2}{4\pi t}}.$$

Osservazione 2.1.5

We can show that a solution exists using the Fourier transform:

$$\begin{cases} \partial_t \hat{u} = -4\pi^2 |\xi|^2 \hat{u} \\ \hat{u}(\xi, \cdot) = \hat{f} \end{cases}.$$

This implies $\hat{u} = e^{-4\pi^2 |\xi|^2 t} \hat{f}$ and so⁵

$$\begin{aligned} u &= (e^{-4\pi^2 |\xi|^2 t})^\vee * f \\ &= \frac{1}{(4\pi t)^{\frac{d}{2}}} e^{-\frac{\pi|x|^2}{4\pi t}} * f \\ &= H_t * f. \end{aligned}$$

Example 2.1.17. If u is the solution of the *heat equation* with initial datum $f \in L^2(\mathbb{R}^d)$ then $H_t * f \rightarrow f$ a.e.x.

Proof. We can see that $\|H_t\|_{L^1} = \|H_1\|_{L^1} = 1$, which means that H_t is scaling invariant. We notice also that $\sup_t \int |H_t|$ and $\forall \delta > 0 \int_{|x|>\delta} |H_t| \xrightarrow{t \rightarrow 0} 0$. Thus H_t is an *approximation of the identity*, i.e.

$$H_t * f \longrightarrow f \quad \text{in } L^2.$$

The pointwise convergence follows because

$$\sup |H_t * f| \leq Mf(x)$$

hence $C = \{f \in L^2 \text{ such that } \lim_{t \rightarrow 0} H_t f = f \text{ almost every } x\}$ is closed. The statement is clearly true for the C_C^∞ which are a dense, so the thesis follows for all $f \in L^2$. $\square$

⁵We consider initial datum $f \in L^2$, so that $\tilde{f} = f$.

Definition 2.1.18 – Schrödinger's kernel

Given the *Schrödinger's equation* with initial datum $g \in L^2(I)$, i.e.:

$$\begin{cases} \partial_t u = i\Delta u \\ u(x) = g \end{cases}$$

we say u is a solution if

$$u = S_t * g,$$

where we define the *Schrödinger's kernel* as

$$S_t = \frac{1}{(4\pi it)^{\frac{d}{2}}} e^{\frac{i\pi|x|^2}{4t}}.$$

Osservazione 2.1.6

We can show that a solution exists using the Fourier transform:

$$\begin{cases} \partial_t \hat{u} = -4\pi^2 i |\xi|^2 \hat{u} \\ \hat{u}(\xi, \cdot) = \hat{g} \end{cases}$$

so $\hat{u} = e^{-4\pi^2 i |\xi|^2 t} \hat{g}$ and so

$$\begin{aligned} u &= (e^{-4\pi^2 i |\xi|^2 t})^\vee * g \\ &= \frac{1}{(4\pi it)^{\frac{d}{2}}} e^{\frac{i\pi|x|^2}{4t}} * g \\ &= S_t * g. \end{aligned}$$

Notice that in this case nor $e^{-4\pi^2 i |\xi|^2 t}$ or u are in L^1 : the equalities follow by direct computations.

Example 2.1.19. If u is the solution of the *Schrödinger's equation* with initial datum $g \in L^2(\mathbb{R}^d)$ it is not true in general that $S_t * g \rightarrow g$ a.e. x .

Proof. The counterexample in $d = 1$ to this was given by König-Dalbert in 1984. If we add some stronger hypothesis, such as⁶ $g \in H^s(\mathbb{R}^d)$ with $s \geq \frac{1}{4}$, thanks to Carleson theorem, we get that $S_t * g \rightarrow g$ a.e. x .

Another difference to the *heat equation* is the following: if we write

$$S_t * g := e^{it\Delta} g$$

then we have that this is a semigroup action on the initial datum, which is also an isometry⁷ on L^2 , i.e.

$$\|e^{it\Delta} g\|_{L^2} = \|g\|_{L^2}.$$

It is also true that

$$\|e^{it\Delta} g\|_{L^\infty} \lesssim \frac{1}{t^{\frac{d}{2}}} \|g\|_{L^1}.$$

⁶by H^s we denote the fractional Sobolev space.

⁷This is not true for the *heat equation*

By the Riesz-Thorin theorem we get the *dispersitive estimate*, i.e.:

$$\|e^{it\Delta}g\|_{L^q} \lesssim \frac{1}{t^{\frac{d}{2}\theta}} \|g\|_{L^p}.$$

The exponents p, q given by the theorem are such that

$$\begin{cases} \frac{1}{p} = \frac{1}{2} + \frac{\theta}{2} \\ \frac{1}{q} = \frac{1}{2} - \frac{\theta}{2} \end{cases} \Rightarrow \frac{1}{p} + \frac{1}{q} = 1.$$

where it is important that $p \in [1, 2]$ and $q \in [2, +\infty]$. Since $\theta = \frac{2}{p} - 1$ we get⁸

$$\begin{aligned} \|e^{it\Delta}g\|_{L^{p'}} &\lesssim \frac{1}{t^{\frac{d}{2}\theta}} \|g\|_{L^p} \\ &= \frac{1}{t^{\frac{d}{p} - \frac{d}{2}}} \|g\|_{L^p} \end{aligned}$$

□

⁸the exponent of t is positive because $p \leq 2$.

2.2 Hardy-Littlewood-Sobolev inequality

Prior to statement and proof of the Hardy- Littlewood-Sobolev inequality, we need the following:

Definition 2.2.1 – Volume and surface of balls

$\nu_d = \mu(\{\|x\|_{\mathbb{R}^d} \leq 1\})$ and ω_{d-1} = surface area.

Proposition 2.2.2

For every d it holds that

$$\omega_{d-1} = \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})} \quad \text{and} \quad \nu_d = \frac{\omega_{d-1}}{d}.$$

Proof. It is an easy computation⁹:

$$\begin{aligned} \pi^{\frac{d}{2}} &= \int_{\mathbb{R}^d} e^{-|x|^2} dx = \omega_{d-1} \int_0^\infty e^{-r^2} r^{d-1} dr \\ &= \frac{\omega_{d-1}}{2} \int_0^\infty e^{-t} t^{\frac{d-1}{2}} t^{-\frac{1}{2}} dt \\ &= \frac{\omega_{d-1}}{2} \Gamma\left(\frac{d}{2}\right). \end{aligned}$$

By definition we have

$$\nu_d = \int_{B(0,1)} 1 dx = \int_0^1 \omega_{d-1} r^{d-1} dr = \frac{\omega_{d-1}}{d}.$$

□

Definition 2.2.3

We define the *multiplicative operator* $|D|^s$ such that:

$$\widehat{|D|^s f} = (2\pi i |\xi|)^s \hat{f}$$

This means^a, if we are allowed, that

$$|D|^s f = ((2\pi i |\xi|)^s \hat{f})^\vee.$$

^aRemembering that $\widehat{\partial_{x_i} f} = 2\pi i \xi_i \hat{f}$, we see, in the sense of *distributions*, that $|D|$ is a derivative operator.

Theorem 2.2.4 – Hardy-Littlewood-Sobolev inequality

For $0 < \gamma < d$ and $1 + \frac{1}{q} = \frac{1}{p} + \frac{\gamma}{d}$, with $1 < p, q < \infty$ we have

$$\left\| \frac{1}{|x|^\gamma} * f \right\|_{L^q} \leq C \|f\|_{L^p} \quad (\text{HLS})$$

⁹We used the following change of variables: $r^2 = t$, $dr = \frac{1}{2} t^{-\frac{1}{2}} dt$.

Osservazione 2.2.1

It is a generalization of Young inequality. Thanks to Example (1.3) we have that $|x|^{-\lambda} \in L^{p,\infty}$ for $p = \frac{d}{\lambda}$, indeed

$$\mu \left\{ \frac{1}{|x|^\gamma} > \lambda \right\} = \mu \left\{ \frac{1}{\lambda} > |x|^\gamma \right\} = \nu_d \frac{1}{\lambda^{\frac{d}{2\gamma}}}.$$

It follows that

$$\left\| \frac{1}{|x|^\gamma} * f \right\|_{L^q} \leq C \left\| \frac{1}{|x|^\gamma} \right\|_{L^{R,\infty}} \|f\|_{L^p}$$

which is like Young inequality with the exponents being

$$1 + \frac{1}{p} = \frac{1}{q} + \frac{\lambda}{d}.$$

Osservazione 2.2.2

Having $(|\xi|^{-k} \hat{f})^\vee = c \int \frac{f(y)}{|x-y|^{d-k}}$ it holds that

$$|D|^{-k} f = \frac{1}{|x|^{\frac{d-k}{\gamma}}} * f.$$

Hence (HLS) becomes:

$$\| |D|^{\gamma-d} f \|_{L^q} \lesssim \|f\|_{L^p}.$$

Osservazione 2.2.3

The exponents p, q given in the statement of (HLS) are the only one possible: we see this thanks to a *scaling argument*. If the inequality is true $\forall f \in L^p$ then it true also for $f_\lambda = f(\frac{x}{\lambda})$, i.e.:

$$\lambda^\smile \| |D|^{\gamma-d} f \|_{L^q} = \| |D|^{\gamma-d} f_\lambda \|_{L^q} \lesssim \|f_\lambda\|_{L^p} = \lambda^{\frac{d}{p}} \|f\|_{L^p}.$$

We need to understand what $\smile$ is: remembering $\partial_{x_i} f_\lambda(x) = \frac{1}{\lambda} \partial_{x_i} f(\frac{x}{\lambda})$ we have that

$$|D|^\alpha f_\lambda(x) = |\lambda|^{-\alpha} |D|^\alpha f\left(\frac{x}{\lambda}\right).$$

This means that

$$\lambda^{-(\gamma-d)+\frac{d}{q}} \|f\|_{L^p} \leq \lambda^{\frac{d}{p}} \|f\|_{L^p}$$

which implies

$$d - \gamma + \frac{d}{q} = \frac{d}{p}.$$

We are finally ready to prove HLS.

Proof of HLS inequality.

$$\begin{aligned} \frac{1}{|x|^\gamma} * f &= \int \frac{1}{|y|^\gamma} f(x-y) dy \\ &= \int_{|y| \leq R} \frac{1}{|y|^\gamma} f(x-y) dy + \int_{|y| > R} \frac{1}{|y|^\gamma} f(x-y) dy \\ &= f * \left(\frac{1}{|y|^\gamma} \chi_{B_R} \right) + \int_{|y| > R} \frac{1}{|y|^\gamma} f(x-y) dy. \end{aligned}$$

We now proceed to estimate separately the two integrals. The first one is an easy computation¹⁰:

$$\begin{aligned}
\left| f * \frac{1}{|y|^\gamma} \chi_{B_R} \right| &\leq Mf \left\| \frac{1}{|y|^\gamma} \chi_{B_R} \right\|_{L^1} \\
&= Mf \int_{B_R} \frac{1}{|y|^\gamma} dy \\
&= Mf \omega_{d-1} \int_0^R \frac{r^{d-1}}{r^\gamma} dr = \frac{\omega_{d-1}}{d-\gamma} R^{d-\gamma} \\
&\lesssim Mf(x) R^{d-\gamma}.
\end{aligned}$$

For the second integral we see that¹¹:

$$\begin{aligned}
\int_{|y|>R} \frac{1}{|y|^\gamma} f(x-y) dy &\leq \|f\|_{L^p} \left\| \frac{1}{|y|^\gamma} \chi_{B_R^c} \right\|_{L^{p'}} \\
&= \|f\|_{L^p} \left(\omega_{d-1} \int_R^\infty \frac{r^{d-1}}{r^{\gamma p'}} dr \right)^{\frac{1}{p'}} \\
&\sim \|f\|_{L^p} R^{\frac{d-\gamma p'}{p'}}
\end{aligned}$$

where we remember that, having $q \neq +\infty$, the is finite¹² since

$$1 < 1 + \frac{1}{q} = \frac{1}{p} + \frac{\gamma}{d} \Rightarrow \frac{\gamma}{d} > \frac{1}{p'}.$$

We proved the following estimate:

$$\left| \frac{1}{|\cdot|^\gamma} * f(x) \right| \lesssim Mf R^{d-\gamma} + \|f\|_{L^p} R^{\frac{d}{p'}-\gamma}$$

Having different exponents we want to find R such that $Mf R^{d-\gamma} = \|f\|_{L^p} R^{\frac{d}{p'}-\gamma}$. It is clear that

$$\frac{Mf}{\|f\|_{L^p}} = R^{\frac{d}{p'}-d} = R^{-\frac{d}{p}} \Rightarrow R = \left(\frac{Mf}{\|f\|_{L^p}} \right)^{-\frac{p}{d}}.$$

With this choice of R we have:

$$\left| \frac{1}{|\cdot|^\gamma} * f(x) \right| \lesssim \|f\|_{L^p}^{1-\frac{p}{q}} Mf(x)^{\frac{p}{q}}.$$

which implies that its L^q norm is:

$$\left\| \frac{1}{|\cdot|^\gamma} * f(x) \right\|_{L^q} \lesssim \|f\|_{L^p}^{1-\frac{p}{q}} \|Mf\|_{L^p}^{\frac{p}{q}}.$$

The Hardy-Littlewood-Sobolev inequality follows since Mf is (p, p) weak, hence

$$\begin{aligned}
\left\| \frac{1}{|\cdot|^\gamma} * f(x) \right\|_{L^q} &\lesssim \|f\|_{L^p}^{1-\frac{p}{q}} \|Mf\|_{L^p}^{\frac{p}{q}} \\
&\lesssim \|f\|_{L^p}^{1-\frac{p}{q}} \|f\|_{L^p}^{\frac{p}{q}} \\
&\lesssim \|f\|_{L^p}.
\end{aligned}$$

We observe that this proof gives a better estimate than the HLS since we have a control with the maximal function. $\square$

¹⁰We stress that it is pointwise.

¹¹We notice that R is elevated to a negative power.

¹²we need to have $d-1-\gamma p' < -1$.

2.2.1 Applications

Thanks to the Hardy-Littlewood-Sobolev inequality we can see if, given the energy E to a problem, we have *coercivity*. By this term we mean the possibility that, if $\|\nabla u\|_{L^p} \rightarrow +\infty$ then $E(u) \rightarrow +\infty$.

Example 2.2.5. Given the problem $-\Delta\phi = |u|^2$, which energy is

$$E(u) = \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|u(x)|^2 |u(y)|^2}{|x - y|} dx dy$$

there exists p such that $|E(u)| \lesssim \|u\|_{L^p}$.

Proof. If we are in $\mathbb{R}^3$ we know the solution to this problem is

$$\Phi = \frac{1}{4\pi} \int \frac{|u(y)|^2}{|x - y|} dy.$$

Remembering that $\int \frac{|u(x)|^2 |u(y)|^2}{|x - y|} = \left(\frac{1}{x} * |u|^2 \right) |u|^2$ then thanks to *HLS* we see that p has to be such that

$$1 + \frac{1}{p'} = \frac{1}{p} + \frac{1}{3} \Rightarrow 2 - \frac{1}{p} = \frac{1}{p} + \frac{1}{3} \Rightarrow p = \frac{6}{5}.$$

This means that

$$\begin{aligned} E(u) &= \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|u(x)|^2 |u(y)|^2}{|x - y|} dx dy = \int_{\mathbb{R}^3} \left(\frac{1}{x} * |u|^2 \right) |u|^2 dx \\ &\leq \|u^2\|_{L^p} \left\| \frac{1}{x} * |u|^2 \right\|_{L^{p'}} \\ &\lesssim \|u\|_{L^p}^2 \|u\|_{L^p}^2 \\ &= \|u^2\|_{L^{\frac{6}{5}}}^2 = \|u\|_{L^{\frac{12}{5}}}^4. \end{aligned}$$

The fact that there is a fourth power should not shock us, as a matter of fact we could immediatly see that, for homogeneity of HLS, we need to have something like $\|u\|_{L^p}^4$, whichever p is. In conclusion we have that, for $p = \frac{12}{5}$ we have

$$|E(u)| \lesssim \|u\|_{L^p}.$$

□

Example 2.2.6. What if we consider the same problem for any dimension d ? The energy in this case is:

$$\iint_{\mathbb{R}^d \times \mathbb{R}^d} \frac{|u(x)|^2 |u(y)|^2}{|x - y|^{d-2}} dx dy.$$

Proof. In this case the solution is, where C is a constant we do not care about, the following:

$$\Phi = C \int \frac{|u(y)|^2}{|x - y|^{d-2}} dy.$$

The exponent given by HLS is

$$1 + \frac{1}{p'} = \frac{1}{p} + \frac{d-2}{d} \Rightarrow p = \frac{2d}{d+2}.$$

So we have

$$\begin{aligned} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \frac{|u(x)|^2 |u(y)|^2}{|x-y|^{d-2}} dx dy &\leq \|u^2\|_{L^p} \|u^2\|_{L^{p'}} \\ &\lesssim \|u^2\|_{L^{\frac{2d}{d+2}}}^2 = \|u\|_{L^{\frac{4d}{d+2}}}^4. \end{aligned}$$

□

We now see whether or not we have *coercivity*: in order to check this we need the Sobolev inequality, stated in 3.1.3, which tells us that in $\mathbb{R}^3$ we have $\|u\|_{L^6} \lesssim \|\nabla u\|_{L^2}$. In general, if we have an energy E we can write $E = K - U$ where K is the kinetic energy and U is the potential one. In this case we want to study

$$\inf_{\substack{\|u\|_{L^2}=1 \\ u \in H^1}} \frac{1}{2} \left\{ \int |\nabla u|^2 - \frac{1}{4} \iint \frac{|u(x)|^2 |u(y)|^2}{|x-y|} dx dy > +\infty \right\}.$$

In this case we have¹³ $K = \int |\nabla u|^2$ and $U = \frac{1}{4} \iint \frac{|u(x)|^2 |u(y)|^2}{|x-y|} dx dy$. We can see that:

$$U \lesssim \|u\|_{L^{\frac{12}{5}}}^4 \lesssim \|u\|_{L^2}^\alpha \|u\|_{L^6}^\beta \lesssim \|u\|_{L^2}^\alpha \|\nabla u\|_{L^2}^\beta.$$

We have *coercivity* if $\beta < 2$. Indeed by Hölder we have

$$\|u\|_{L^p} \leq \|u\|_{L^2}^\theta \|u\|_{L^6}^{1-\theta}.$$

where

$$\frac{1}{p} = \frac{\theta}{2} + \frac{1-\theta}{6} = \frac{3\theta+1-\theta}{6} \Rightarrow \frac{2\theta}{6} = \frac{1}{p} - \frac{1}{6} = \frac{6-p}{6p} \Rightarrow \theta = \frac{6-p}{2p}.$$

If $p = \frac{12}{5}$ we get $\theta = \frac{3}{4}$, hence

$$U \lesssim \|u\|_{L^2}^3 \|\nabla u\|_{L^2}.$$

□

Example 2.2.7. Given the problem^a $i\partial_t u = -\Delta u - \frac{1}{|x|}u$, prove that

$$\inf_{\|u\|_{L^2}=1} \frac{1}{2} \int |\nabla u|^2 dx - \frac{1}{2} \int \frac{|u|^2}{|x|} dx > -\infty.$$

^aIt arises from quantum mechanics

Proof. Using the Hardy inequality (3.1.1):

$$\left(\int_{\mathbb{R}^d} \frac{|u|^2}{|x|^2} dx \right)^{\frac{1}{2}} \leq \frac{2}{d-2} \|\nabla u\|_{L^2}$$

we have $\inf > -\infty$. □

¹³Physically this are the kinetic and potential energy associated to an electric charge

Chapter 3

The Hilbert and Riesz Transform

3.1 Schwartz class and Distributions

Thanks to Plancherel equality and Riesz-Thorin theorem we are able to define the Fourier transform for functions $f \in L^p$ with $1 \leq p \leq 2$. The aim of this chapter is to generalize this definition, in order to have a Fourier transform for more objects, which will be the *tempered distributions*.

3.1.1 Schwartz class of functions

The idea will be, given some linear operator T acting on a class of function $\mathcal{S}$, to define $\hat{T}$ such that¹

$$\langle \hat{T}, u \rangle = \langle T, \hat{u} \rangle.$$

It is immediate that we need to define a class of function $\mathcal{S}$ such that, if $u \in \mathcal{S}$, also $\hat{u} \in \mathcal{S}$.

Let us now fix some notation²: given $x \in \mathbb{R}^d$ and $\alpha = (\alpha_1, \dots, \alpha_d)$ a multiindex, where $\alpha_i \geq 0 \forall i$ and $|\alpha| = \sum_{i=1}^d \alpha_i$, we write

$$x^\alpha = x_1^{\alpha_1} \dots x_d^{\alpha_d}$$

$$\partial^\alpha f = \partial_{x_1}^{\alpha_1} \dots \partial_{x_d}^{\alpha_d} f$$

It is immediate to observe that:

Proposition 3.1.1

For every $x \in \mathbb{R}^d$ and α multiindex the following hold:

1.

$$|x|^\alpha \leq C_{d,\alpha} |x|^{|\alpha|}.$$

2. For all $k \in \mathbb{Z}_+$

$$|x|^k \leq C_{k,d} \sum_{|\alpha|=k} |x^\alpha|$$

Proof. This inequalities follows in the same way: let us considere the following maps:

$$\begin{array}{ccc} \xi_1 : \mathbb{S}^{d-1} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & |x^\alpha| \end{array}.$$

¹In this case by $\langle T, u \rangle$ we mean the evaluation of T in u.

²we will use the same notation as in Grafakos

$$\begin{aligned} \xi_2 : \mathbb{S}^{d-1} &\longrightarrow \mathbb{R} \\ x &\longmapsto \sum_{|\alpha|=k} |x^\alpha| . \end{aligned}$$

Both of them have the minimum which is greater than 0 for construction, by homogeneity we get the thesis. $\square$

Osservazione 3.1.1

In the first inequality, it is not possible to have two different multiindexes α, β :

$$|x^\alpha| \preccurlyeq |x|^{|\beta|}.$$

In order to see this it is sufficient to do a *rescaling argument*: if it is true for x it should also hold for λx , where $\lambda \in \mathbb{R}$.

Proposition 3.1.2 – Leibnitz rule

For all $f, g \in \mathbb{R}^d$ we have

$$\partial^\alpha (fg) = \sum_{\beta \leq \alpha} \binom{\alpha_1}{\beta_1} \cdots \binom{\alpha_d}{\beta_d} \partial^\beta f \partial^{\alpha-\beta} g$$

where $\beta \leq \alpha$ if $\beta_j \leq \alpha_j \forall 1 \leq j \leq d$.

Definition 3.1.3 – Schwartz function

$f \in C^\infty$ is in the Schwartz class $\mathcal{S}$ if $\forall \alpha, \beta$ multiindex we have

$$\rho_{\alpha, \beta}(f) = \sup_{x \in \mathbb{R}^d} |x^\alpha \partial^\beta f| < +\infty.$$

This will be a class of test functions.

Osservazione 3.1.2

We will often use this equivalent definition: given $f \in C^\infty$ the following holds:

$$f \in \mathcal{S} \iff |\partial^\beta f(x)| \leq C_{\beta, N} \frac{1}{1 + |x|^N} \quad \forall \beta \forall N.$$

This means that all the derivatives of Schwartz functions decay faster than every polynomial. From this equivalent definition is trivial to deduce that $\mathcal{S} \subseteq L^p$ for every $p \geq 1$.

Proposition 3.1.4

It is true that

1. if $f \in C_c^\infty$ then $f \in \mathcal{S}$
2. if p is a polynomial and $f \in \mathcal{S}$ then $p \cdot f \in \mathcal{S}$
3. if $f \in \mathcal{S}$ then for every α, β then $\partial^\alpha f \in \mathcal{S}$ and $x^\alpha \partial^\beta f \in \mathcal{S}$

Definition 3.1.5 – Convergence in Schwartz's class

We say that given $f_n \in \mathcal{S}$, $f \in \mathcal{S}$ then $f_n \xrightarrow{\mathcal{S}} f$ if $\forall \alpha, \beta$

$$\rho_{\alpha, \beta}(f_n - f) \xrightarrow{n \rightarrow \infty} 0.$$

Proposition 3.1.6

For $p \in [1, +\infty]$ if $f_n \xrightarrow{\mathcal{S}} f$ then^a $f_n \xrightarrow{L^p} f$ and

$$\|\partial^\beta f\|_{L^p} \leq C_{d,p} \sum_{|\alpha| \leq \left\lfloor \frac{d+1}{p} \right\rfloor + 1} \rho_{\alpha, \beta}(f).$$

^aThis implies that the convergence in $\mathcal{S}$ is stronger than convergence in all L^p .

Proof. Let us prove that, given the estimate, then the $\mathcal{S}$ -convergence implies the L^p -convergence:

$$\|f_n - f\|_{L^p} \preccurlyeq \sum_{|\alpha| \leq \left\lfloor \frac{d+1}{p} \right\rfloor + 1} \rho_{\alpha, 0}(f_n - f) \rightarrow 0$$

because is a finite sum of numbers going to 0.

Let us prove the estimate:

$$\begin{aligned} \|\partial^\beta f\|_{L^p} &= \left(\int_{\mathbb{R}^d} |\partial^\beta f|^p dx \right)^{\frac{1}{p}} \\ &= \left(\int_{|x| < 1} |\partial^\beta f|^p dx + \int_{|x| \geq 1} |\partial^\beta f|^p dx \right)^{\frac{1}{p}} \\ &\leq \left(\nu_d \left(\sup_{x \in \mathbb{R}^d} |\partial^\beta f| \right)^p + C \sup_{x \geq 1} |x|^{d+1} |\partial^\beta f|^p \right)^{\frac{1}{p}} \\ &\leq \left(\nu_d \left(\sup_{x \in \mathbb{R}^d} |\partial^\beta f| \right)^p + C \sup_{x \geq 1} |x|^{\left\lfloor \frac{d+1}{p} \right\rfloor + 1} |\partial^\beta f|^p \right)^{\frac{1}{p}} \\ &\preccurlyeq \rho_{0, \beta}(f) + \sum_{|\alpha| = \left\lfloor \frac{d+1}{p} \right\rfloor + 1} \rho_{\alpha, \beta}(f) \end{aligned}$$

where the last inequality follows from the property of multiindexes and where we also used the fact that

$$\int_{|x| \geq 1} |\partial^\beta f|^p dx = \int_{|x| \geq 1} |x|^{d+1} |x|^{-d-1} |\partial^\beta f|^p dx \leq C \sup_{x \geq 1} |x|^{d+1} |\partial^\beta f|^p.$$

□

We can see that the $\mathcal{S}$ class is closed under many operations such as convolution and Fourier transform.

Proposition 3.1.7

Given $f, g \in \mathcal{S}$ we have that $f * g \in \mathcal{S}$.

Proof. For any $x, y \in \mathbb{R}^d$ and $N \in \mathbb{N}$ we have:

$$\frac{1}{(1 + |x - y|)^N} \leq \frac{(1 + |y|)^N}{(1 + |x|)^N}.$$

It is sufficient to prove this for $N = 1$ and then it will follow for induction. When $N = 1$ we can see that

$$1 + |x| = 1 + |x - y + y| \leq 1 + |x - y| + |y| \leq (1 + |x - y|)(1 + |y|).$$

The induction part of the proof is left to the reader.

If we now compute $f * g(x)$ we have:

$$\begin{aligned} |f * g| &= \left| \int f(x - y)g(y)dy \right| \\ &\leq \int \frac{1}{(1 + |x - y|)^N} \frac{1}{(1 + |y|)^{N+d+1}} dy \\ &\leq \frac{1}{(1 + |x|)^N} \int \frac{(1 + |y|)^N}{(1 + |y|)^{N+d+1}} dy \\ &= \frac{1}{(1 + |x|)^N} \int \frac{1}{(1 + |y|)^{d+1}} < +\infty. \end{aligned}$$

We now have to check all the derivatives $\partial^\beta(f * g)$, but this is trivial since:

$$\partial^\beta(f * g) = \partial^\beta f * g.$$

□

Lemma 3.1.8

For any $f \in \mathcal{S}$ and any α multiindex we have:

1. $\widehat{\partial^\alpha f} = (2\pi i \xi)^\alpha \hat{f}(\xi)$
2. $\partial^\alpha \hat{f}(\xi) = (-2\pi i x)^\alpha f(\xi)$

Proof. For simplicity we will assume we only have one derivative, i.e. $\alpha = (1, 0, \dots, 0)$.

Proof of 1):

$$\begin{aligned} \widehat{\partial^\alpha f} &= \int_{\mathbb{R}^d} \partial^\alpha f(x) e^{-2\pi i \xi \cdot x} dx \\ &= (-1)^{|\alpha|} \int (-2\pi i \xi)^\alpha f(x) e^{-2\pi i \xi \cdot x} dx \\ &= (2\pi i \xi)^\alpha \hat{f}(\xi). \end{aligned}$$

The integration by parts is possible because, considering $R \rightarrow +\infty$, the following holds:

$$\int_{B_R} \partial_{x_1} f e^{-2\pi i \xi \cdot x} dx = - \int_{B_R} f (-2\pi i \xi_1) e^{-2\pi i \xi \cdot x} dx + \int_{\partial B_R} f e^{-2\pi i \xi \cdot x} \nu_1 d\sigma$$

Idea This trick is quite common for Schwartz functions which decay really fast. We want to see that the second term goes to zero so it's done: it is quite obvious because f decays faster than any polynomial. $\lrcorner$

Proof of 2):

$$\begin{aligned}
 \partial_{\xi_1} \hat{f}(\xi) &= \lim_{h \rightarrow 0} \frac{\hat{f}(\xi + h e_1) - \hat{f}(\xi)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \int f(x) \left[e^{-2\pi i(\xi + h e_1) \cdot x} - e^{-2\pi i \xi \cdot x} \right] dx \\
 &= \lim_{h \rightarrow 0} \int f(x) e^{-2\pi i \xi \cdot x} \frac{e^{-2\pi i(h x_1)} - 1}{h} dx \\
 &= \int f(x) e^{-2\pi i \xi \cdot x} \lim_{h \rightarrow 0} \frac{e^{-2\pi i(h x_1)} - 1}{h} dx \\
 &= (-2\pi i x_1)^\alpha f.
 \end{aligned}$$

where the limit passed inside because

$$e^{-2\pi i \xi \cdot x} \frac{e^{-2\pi i(h x_1)} - 1}{h} - (-2\pi i x_1) e^{-2\pi i \xi \cdot x} = H(x)$$

goes to 0 as $h \rightarrow 0$. Moreover, since $|H| \leq C|x|$ and $|x|$ is L^1 function with the measure $d\mu = f(x)dx$ (since $f \in \mathcal{S}$), we can use the Dominated Convergence Theorem. $\square$

Theorem 3.1.9

If $f \in \mathcal{S}$ then $\hat{f} \in \mathcal{S}$.

Proof. We want to see that $x^\alpha \partial^\beta \hat{f}$ in L^∞ for any α, β . The idea is to see that something like this holds:

$$\|x^\alpha \partial^\beta \hat{f}\|_{L^\infty} = \|\hat{\smile}\|_{L^\infty} \leq \|\smile\|_{L^1}$$

Let us understand what the $\smile$ is: by the previous formulas we have

$$\|x^\alpha \partial^\beta \hat{f}\|_{L^\infty} = \|\widehat{\partial^\alpha (x^\beta f)}\|_{L^\infty} \frac{(2\pi)^{|\beta|}}{2\pi^{|\alpha|}} \leq C \|\partial^\alpha (x^\beta f)\|_{L^1} < +\infty$$

which is finite since $f \in \mathcal{S} \Rightarrow f \in L^1$ and $f \in \mathcal{S} \Rightarrow \partial^\alpha (x^\beta f) \in \mathcal{S}$. $\square$

Example 3.1.10. For $d = 1$ let $g = \chi_{[a,b]}(x)$. We want to compute $\hat{g}$.

$$\hat{g}(\xi) = \int_a^b g(x) e^{-2\pi i \xi x} dx = \left[\frac{e^{-2\pi i \xi x}}{-2\pi i \xi} \right]_a^b = \frac{e^{-2\pi i \xi a} - e^{-2\pi i \xi b}}{2\pi i \xi}.$$

Example 3.1.11. For any d let us consider $g = \prod_{i=1}^d \chi_{[a_i, b_i]}(x_i)$. So

$$\hat{g}(\xi) = \prod_{i=1}^d \left[\frac{e^{-2\pi i \xi_i a_i} - e^{-2\pi i \xi_i b_i}}{2\pi i \xi_i} \right].$$

Is it true that $\hat{g}$ goes like $\frac{1}{|x|}$? We see that if $\xi \in \mathbb{R}^d \setminus \{0\}$ then there exists i_0 such that $|\xi_{i_0}| > \frac{|\xi|}{\sqrt{d}}$. If this were not the case we would have

$$|\xi|^2 = \sum x_i^2 < \sum \frac{|\xi|^2}{d} = |\xi|^2$$

which is absurd. So in the previous computation we have:

$$|\hat{g}(\xi)| \leq \frac{2}{2\pi} \frac{\sqrt{d}}{|\xi|} \prod_{i \neq i_0} (b_i - a_i)$$

where we used the Lagrange's estimate to write $\frac{e^{-2\pi i \xi_i a_i} - e^{-2\pi i \xi_i b_i}}{2\pi i \xi_i} \leq (b_i - a_i)$.

Theorem 3.1.12

If $f \in L^1$ we have $\hat{f}(\xi) \rightarrow 0$ for $|\xi| \rightarrow +\infty$.

Proof. We know that for each $f \in L^1$ there exist g simple function such that, for $\varepsilon > 0$, $\|f - g\|_{L^1} \leq \frac{\varepsilon}{2}$. We see that

$$|\hat{f}(\xi)| = |\hat{f}(\xi) - \hat{g}(\xi) + \hat{g}(\xi)| \leq |\hat{f}(\xi) - \hat{g}(\xi)| + |\hat{g}(\xi)| \leq \|f - g\|_{L^1} + \frac{C}{|\xi|}.$$

The thesis follows because for $|\xi| > M_0$ we have $|\hat{g}| \leq \frac{\varepsilon}{2}$, which gives $|\hat{f}(\xi)| \leq \varepsilon$. $\square$

Proposition 3.1.13 – Hardy inequality

Given $d \geq 3$ for all $f \in C_c^\infty(\mathbb{R}^d \setminus \{0\})$ we have:

$$\left(\int_{\mathbb{R}^d} \frac{|f(x)|^2}{|x|^2} dx \right)^{\frac{1}{2}} \leq \frac{2}{d-2} \|\nabla f\|_{L^2}.$$

Proof of Hardy. The best constant is never achieved.

The reason why we ask $f \in C_c^\infty(\mathbb{R}^d \setminus \{0\})$ is that we don't want problems when integrating by parts. Remember³

$$\frac{1}{|x|^2} = -\frac{1}{2} x \cdot \nabla \left(\frac{1}{|x|^2} \right)$$

So if we define $R(f) = \sum x_i \frac{\partial f}{\partial x_i}$ we have the following properties:

$$\begin{aligned} \frac{1}{|x|^2} &= -\frac{1}{2} R \left(\frac{1}{|x|^2} \right) \\ |R(f)|^2 &= \left| \sum x_i \partial_{x_i} (f) \right|^2 \leq \sum_{i=1}^d x_i^2 \sum_{i=1}^d |\partial_{x_i} f|^2 = \sum x_i^2 |\nabla f|^2. \end{aligned}$$

³Since $\partial_{x_i} \frac{1}{|x|^2} = \frac{-2}{|x|^3} \frac{x_i}{|x|} = \frac{-2x_i}{|x|^4}$ then $\sum x_i \cdot \frac{-2x_i}{|x|^4} = \frac{-2}{|x|^2}$.

Let's prove the inequality:

$$\begin{aligned}
 \int_{\mathbb{R}^d} \frac{|f(x)|^2}{|x|^2} dx &= \int |f|^2 \left(-\frac{1}{2} x \cdot \nabla \left(\frac{1}{|x|^2} \right) \right) dx \\
 &= \frac{1}{2} \int \frac{1}{|x|^2} \sum_{i=1}^d \partial_{x_i} (x_i |f|^2) dx \\
 &= \frac{d}{2} \int \frac{|f|^2}{|x|^2} + \frac{1}{2} \int \frac{1}{|x|^2} \sum x_i [f_{x_i} \bar{f} + f \bar{f}_{x_i}] dx
 \end{aligned}$$

where we use the fact that, having $|f|^2 = f \bar{f}$, it follows that $\partial_{x_i} (f \bar{f}) = f_{x_i} \bar{f} + f \bar{f}_{x_i}$. Until now we have proved that:

$$\frac{2-d}{2} \int \frac{|f|^2}{|x|^2} = \frac{1}{2} \int \frac{R(f) \bar{f}}{|x|^2} + \frac{R(\bar{f}) f}{|x|^2} dx.$$

By applying Cauchy-Schwartz we get

$$\int \left| \frac{R(f) \bar{f}}{|x|^2} \right| dx = \left(\int \frac{|R(f)|^2}{|x|^2} \right)^{\frac{1}{2}} \left(\int \frac{|\bar{f}|^2}{|x|^2} \right)^{\frac{1}{2}}$$

and in conclusion

$$\begin{aligned}
 \frac{d-2}{2} \left(\int \frac{|f|^2}{|x|^2} \right)^{\frac{1}{2}} &\leq \left(\int \frac{|R(f)|^2}{|x|^2} dx \right)^{\frac{1}{2}} \\
 &\leq \left(\int |\nabla f|^2 dx \right)^{\frac{1}{2}}.
 \end{aligned}$$

□

Osservazione 3.1.3

From this inequality we have:

$$\int \frac{|f|^2}{|x|^2} dx \preccurlyeq \|\nabla f\|_{L^2}^2 = \| |\xi| \hat{f} \|_{L^2}^2.$$

It is possible to generalize and get, $\forall s, 0 < s < \frac{d}{2}$

$$\int \frac{|f|^2}{|x|^{2s}} \preccurlyeq \| |\xi|^s \hat{f} \|_{L^2}^2.$$

Proposition 3.1.14 – Heisenberg's inequality

For any $f \in \mathcal{S}$ we have

$$\|f\|_{L^2}^2 \leq \frac{4\pi}{d} \inf_{y \in \mathbb{R}^d} \left(\int |x-y|^2 |f(x)|^2 dx \right)^{\frac{1}{2}} \inf_{z \in \mathbb{R}^d} \left(\int |\xi-z|^2 |\hat{f}(\xi)|^2 d\xi \right)^{\frac{1}{2}}$$

Osservazione 3.1.4

In some books you can find a different statement, which usually is

$$\|f\|_{L^2}^2 \leq C \|xf\|_{L^2} \|\nabla f\|_{L^2}.$$

The problem with this formulation is that, its *physical meaning* is not really clear. If you consider f as the position of a particle and $\hat{f}$ as its frequency, you cannot have a peak for both of them because, if this were the case, you would have both of the $\inf = 0$. Let us remember then having a peak means that we know with certainty the quantity represented by those functions.

Proof of Heisenberg. Let us see it for $d = 1$.

$$\|f\|_{L^2}^2 = \int f \bar{f} \partial_x(x - y) dx = - \int \partial_x(f \bar{f})(x - y) dx = - \int (f_x \bar{f} + f \bar{f}_x)(x - y) dx$$

By cauchy schwartz

$$\|f\|_{L^2}^2 \leq 2 \left(\int |f_x|^2 dx \right)^{\frac{1}{2}} \left(\int |\bar{f}|^2 dx \right)^{\frac{1}{2}} = (*)$$

and since

$$\left(\int |f_x|^2 dx \right)^{\frac{1}{2}} = \left(\int 4\pi^2 |\xi|^2 |\hat{f}|^2 d\xi \right)^{\frac{1}{2}}$$

we get

$$(*) = 4\pi \left(\int |\xi|^2 |\hat{f}(\xi)|^2 d\xi \right)^{\frac{1}{2}} \left(\int |f|^2(x - y)^2 \right)^{\frac{1}{2}}.$$

since the LHS doesn't depend upon y we can take the $\inf$ on y . Since $\widehat{f e^{-2\pi i x z}} = \hat{f}(\xi - z)$ the same steps can be run over with $\hat{f}$ in order to complete the proof. $\square$

3.1.2 Distributions

The idea behind the theory of distribution is that sometimes working with “bad” functions can be harder then working with linear functionals acting on “nice” functions.

Notation: $C_0^\infty = \{\text{smooth function with compact support}\}$. We have $C_0^\infty \subset \mathcal{S} \subset C^\infty$. These three spaces of test functions will generate three dual spaces:

$$(\mathcal{S})' = \mathcal{S}' = \{\text{tempered distributions}\}$$

$$(C_0^\infty)' = D' = \{\text{distributions}\}$$

$$(C^\infty)' = \mathcal{E}' = \{\text{distribution with compact support}\}$$

Taking the dual reverses the inclusions, hence $\mathcal{E}' \subset \mathcal{S}' \subset D'$. We recall what the convergence in these spaces is:

- $f_k \xrightarrow{\mathcal{S}} f$ if $\forall \alpha, \beta \rho_{\alpha, \beta}(f_k - f) \rightarrow 0$.
- $f_k \xrightarrow{C_0^\infty} f$ if $\forall \alpha$ we have $\|\partial^\alpha(f_k - f)\|_{L^\infty} \rightarrow 0$ where $\text{supp } f_k \subset B$ for all k with B compact.
- $f_k \xrightarrow{C^\infty} f$ if $\forall \alpha, \forall N > 0$ we have $\sup_{|x| \leq N} |\partial^\alpha f_k - f| \rightarrow 0$.

Taking $X \in \{C^\infty, \mathcal{S}, C_0^\infty\}$ we say that $(T_k)_k \subset X'$ converges to $T \in X'$ if

$$T_k(f) \longrightarrow T(f)$$

for every $f \in X$.

Example 3.1.15. For any $\varphi \in C_0^\infty$ with $d = 1$ we consider $\varphi_k = \frac{1}{k}\varphi(x - k)$. We can see that $\varphi_k \xrightarrow{C^\infty} 0$ whereas $\varphi_k \not\xrightarrow{\mathcal{S}} 0$. The first one follows because φ has compact support. Let's now see why we do not have a convergence in the Schwartz class. Assume that it converges to 0, then we have

$$\rho_{1,0} = \sup_{x \in \mathbb{R}} \left| \frac{x\varphi(x - k)}{k} - 0 \right|$$

Claim is that $\rho_{1,0} \xrightarrow{k \rightarrow +\infty} 0$, if we consider $x = k$ then we have $\varphi(0)$ which can be whatever we want, so absurd.

Proposition 3.1.16

A linear functional T acting on $\mathcal{S}$, so $T : \mathcal{S} \rightarrow \mathbb{C}$, is a *tempered distribution* if and only if there exists m, k such that

$$|\langle T, f \rangle| \leq c \sum_{\substack{|\alpha| \leq m \\ |\beta| \leq k}} \rho_{\alpha,\beta}(f) \quad \forall f \in \mathcal{S}$$

Proof. Omitted. □

Example 3.1.17 (Dirac mass at zero). We denote the dirac mass at the point a with δ_a . We define δ_a as the operator such that

$$\langle \delta_a, \varphi \rangle = \varphi(a) \quad \forall \varphi \in C^\infty.$$

We have $\delta_0 \in \mathcal{E}'$, indeed if we have $\varphi_k \xrightarrow{C^\infty} \varphi$ then

$$\langle \delta_0, \varphi_k \rangle \rightarrow \langle \delta_0, \varphi \rangle$$

Because $\mathcal{E}' \subset \mathcal{S}' \subset D'$, it is true that $\delta_0 \in \mathcal{S}$ and $\delta_0 \in D'$.

Idea Different spaces are used in different situations. For instance in Harmonic Analysis we will mostly use $\mathcal{S}'$ instead in PDE Theory D' is the obvious choice. ┘

Example 3.1.18. if $f \in L^p$ with $1 \leq p \leq +\infty$ then if we consider

$$\langle T_f, \varphi \rangle = T_f(\varphi) = \int_{\mathbb{R}^d} f\varphi dx$$

we have $T_f \in D'$.

$$|\langle T_f, \varphi \rangle| \leq \|\varphi\|_{L^{p'}} \|f\|_{L^p}$$

so if $\varphi_k \xrightarrow{\mathcal{S}} \varphi$ then $\varphi_k \xrightarrow{L^p} \varphi$ and

$$|\langle T_f, \varphi_j - \varphi \rangle| \leq \|\varphi_j - \varphi\|_{L^{p'}} \|f\|_{L^p}.$$

Example 3.1.19. If we have $|g| \lesssim (1 + |x|)^k$ for some k , where g is a measurable function

$$\begin{aligned} \left| \int_{\mathbb{R}^d} g \varphi dx \right| &= |T_g(\varphi)| \lesssim \int_{\mathbb{R}^d} (1 + |x|)^k |\varphi(x)| dx \\ &= \int_{\mathbb{R}^d} (1 + |x|)^m (1 + |x|)^{k-m} |\varphi(x)| dx \\ &\stackrel{*}{\lesssim} \sup_{x \in \mathbb{R}^d} (1 + |x|)^m |\varphi(x)| \int_{\mathbb{R}^d} (1 + |x|)^{k-m} dx \end{aligned}$$

where $*$ follows from $\varphi \in \mathcal{S}$. If $k - m < d$ the function is integrable. So we take m such that $m > k - d$ and we have that $T_g(\varphi)$ is controlled by a finite number of seminorms, so it is a tempered distribution.

We see now differentiation of distributions and the Fourier transform of tempered distribution.

Definition 3.1.20 – Differentiation of distribution

We define^a for $\varphi \in C_0^\infty$

$$\langle \partial^\alpha T, \varphi \rangle = (-1)^{|\alpha|} \langle T, \partial^\alpha \varphi \rangle$$

^athe term $(-1)^{|\alpha|}$ is necessary in order to respect the integration by parts.

Definition 3.1.21 – Fourier Transform and Antitransform of tempered distribution

Given $\varphi \in \mathcal{S}$ then^a

$$\langle \hat{T}, \varphi \rangle = \langle T, \hat{\varphi} \rangle.$$

We also have

$$\langle \check{T}, \varphi \rangle = \langle T, \check{\varphi} \rangle$$

^aIt is well defined since $f \in \mathcal{S} \Rightarrow \hat{f} \in \mathcal{S}$.

Proposition 3.1.22

It holds

$$\widehat{\partial^\alpha \delta_0} = (2\pi i x)^\alpha.$$

Proof. We want to see that

$$\langle \widehat{\partial^\alpha \delta_0}, \varphi \rangle = \int (2\pi i x)^\alpha \varphi dx$$

Idea Distributions *act* on functions. So you consider the integral. ┘

$$\begin{aligned}
\langle \widehat{\partial^\alpha \delta_0}, \varphi \rangle &= \langle \partial^\alpha \delta_0, \hat{\varphi} \rangle \\
&= (-1)^{|\alpha|} \langle \delta_0, \partial^\alpha \hat{\varphi} \rangle \\
&= (-1)^{|\alpha|} \langle \delta_0, \widehat{(-2\pi i x)^\alpha \varphi} \rangle \\
&= \langle \delta_0, \widehat{(2\pi i x)^\alpha \varphi} \rangle \\
&= \widehat{(2\pi i x)^\alpha \varphi}(0) = \int (2\pi i x)^\alpha \varphi(x) e^{-2\pi i 0 \cdot x}.
\end{aligned}$$

In particular $\widehat{\delta_0} = 1$. □

Idea There are distributions whose Fourier transform are functions. So the next definition should not impress ourselves: ┘

We can consider the homogeneous Sobolev space

$$\dot{H}^s = \{\varphi \in \mathcal{S}', \hat{\varphi} \in L_{loc}^1 \text{ s.t. } \int |\xi|^{2s} |\hat{\varphi}|^2 d\xi < +\infty\}.$$

Esercizio 3.1.1

Prove that

$$\widehat{\delta_a} = e^{-2\pi i a \cdot x}$$

3.1.3 Sobolev Inequality

We are set on $\mathbb{R}^d$. We consider, given $\varphi \in \mathcal{S}$, the multiplicative operator $|D|$ as in 2.2

$$\widehat{|D|^s \varphi} = (2\pi i |\xi|)^s \hat{\varphi}.$$

Osservazione 3.1.5

Recalling that $\widehat{\partial_{x_j} \varphi} = 2\pi i \xi_j \hat{\varphi}$ and $\widehat{\Delta \varphi} = -4\pi^2 |\xi|^2 \hat{\varphi}$ then

$$|D|^2 = -\Delta$$

Theorem 3.1.23 – Sobolev Inequality

For $1 < p < +\infty$ and $f \in \mathcal{S}$ such that $|D|^s f \in L^p$ then

$$\|f\|_{L^q} \preccurlyeq \| |D|^s f \|_{L^p}$$

where $\frac{1}{p} = \frac{1}{q} + \frac{s}{d}$.

Osservazione 3.1.6

The noble example, the one really common to use, is for $p = 2$ and $s = 1$. Assuming $d \geq 3$ we get $q = \frac{2d}{d-2}$. The Sobolev inequality takes the form

$$\|f\|_{L^{\frac{2d}{d-2}}} \leq \| |D| f \|_{L^2}$$

By Plancherel we also have that

$$\| |D|f \|_{L^2} = \| \nabla f \|_{L^2}$$

since

$$\| \widehat{|D|f} \|_{L^2}^2 = \| \widehat{\nabla f} \|_{L^2}^2$$

and

$$\int \| \widehat{|D|f} \|^2 = \int 4\pi^2 |\xi|^2 |\hat{f}|^2 = \int |\nabla f|^2.$$

In general for any $p \neq 2$, where we don't have Plancherel, we have an estimate with the Hilbert/Riesz⁴ transform which gives

$$\| |D|u \|_{L^p} \lesssim \| \nabla u \|_{L^p}$$

As always, let us see that the exponents have to be those by a scaling argument: by defining $f_\lambda = f(\frac{x}{\lambda})$ we get

$$\| f_\lambda \|_{L^q} = \lambda^{\frac{d}{q}} \| f \|_{L^q}$$

and

$$|D|^s f_\lambda = \lambda^{-s} |D|^s f \left(\frac{x}{\lambda} \right)$$

so

$$\| |D|^s f_\lambda \|_{L^p} = \lambda^{-s+\frac{d}{p}} \| |D|^s f \|_{L^p}$$

from which the right condition is

$$\frac{d}{q} = -s + \frac{d}{p}.$$

Proof of Sobolev's theorem. We recall that⁵

$$\| f \|_{L^q} = \sup_{\substack{\|g\|_{L^{p'}}=1 \\ g \in \mathcal{S}}} |\langle f, g \rangle|$$

and then, thanks to Plancherel equality

$$= \sup_{\substack{\|g\|_{L^{p'}}=1 \\ g \in \mathcal{S}}} |\langle \hat{f}, \hat{g} \rangle| = \sup_{\substack{\|g\|_{L^{p'}}=1 \\ g \in \mathcal{S}}} |\langle (2\pi|\xi|)^s \hat{f}, (2\pi|\xi|)^{-s} \hat{g} \rangle|.$$

Since $(2\pi|\xi|)^s \hat{f}$ is a tempered distribution (see 3.1.1) we just need to prove that $(2\pi|\xi|)^{-s} \hat{g} \in \mathcal{S}$. This is true if $\hat{g} = 0$ around the origin.

Idea The idea is to consider $F \subset \mathcal{S}$ as the Schwartz function with this property and show they are dense in $\mathcal{S}$. We can then take the supremum on this subset. $\lrcorner$

If this were the case we would have

$$\sup_{g \in F} |\langle |D|^s f, |D|^{-s} g \rangle|$$

⁴The reader should not be worried, they will come soon.

⁵We recall that $\mathcal{S}$ is dense in $L^{p'}$.

where $F = \{g \in \mathcal{S}, \hat{g} = 0 \text{ in a neighborhood of the origin}\}$. So

$$\|f\|_{L^q} = \sup_{\substack{g \in F \\ \|g\|_{L^{p'}=1}} \langle |D|^s f, |D|^{-s} g \rangle \leq \sup_{\substack{g \in F \\ \|g\|_{L^{p'}=1}} \| |D|^s f \|_{L^p} \| |D|^{-s} g \|_{L^{p'}}$$

(we applied Hölder). In order to conclude we use *HLS* to see that

$$\| |D|^{-s} g \|_{L^{p'}} \preccurlyeq \|g\|_{L^{q'}} = 1$$

How do we apply *HLS*? If $0 < s < d$

$$|D|^{-s} g = c \frac{1}{|x|^{d-s}} * g$$

and we want to see

$$\left\| \frac{1}{|x|^{d-s}} * g \right\|_{L^{p'}} \preccurlyeq \|g\|_{L^{q'}}$$

Remembering

$$\left\| \frac{1}{|x|^\alpha} * g \right\|_{L^{p'}} \preccurlyeq \|g\|_{L^{q'}}$$

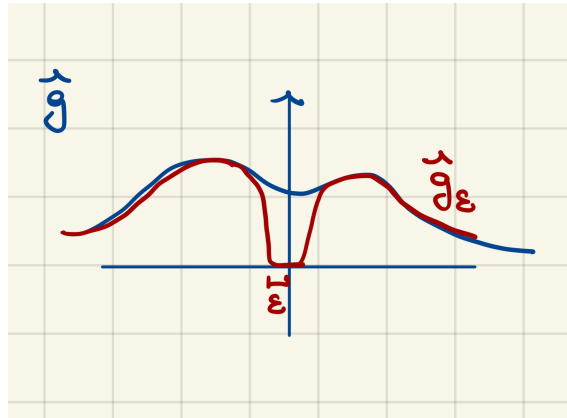
where

$$1 + \frac{1}{p'} = \frac{1}{q'} + \frac{\alpha}{d}$$

with our α we have $\frac{1}{p'} = \frac{1}{q'} + \frac{s}{d}$.

The proof is complete once we show that F is dense in $\mathcal{S}$. Let us consider $g \in \mathcal{S}$ and a bump function $\varphi(t) = \begin{cases} 1 & |t| \leq 1 \\ 0 & |t| > 2 \end{cases}$. Define

$$\hat{g}_\varepsilon(\xi) = \hat{g}(\xi) \left[1 - \varphi\left(\frac{|\xi|}{\varepsilon}\right) \right].$$



Clearly $\hat{g}_\varepsilon \in F$. So we just have to prove there is convergence in L^p :

$$g - g_\varepsilon = \left(\hat{g} \varphi\left(\frac{|\xi|}{\varepsilon}\right) \right)^\vee = \varepsilon^d g * \check{\varphi}(\varepsilon \cdot)(x)$$

So⁶

$$\|g - g_\varepsilon\|_{L^p} = \varepsilon^d \|g * \check{\varphi}(\varepsilon \cdot)(x)\|_{L^p} \leq \varepsilon^d \|g\|_{L^1} \varepsilon^{-\frac{d}{p}} \|\check{\varphi}\|_{L^p} = \varepsilon^{d(1-\frac{1}{p})} \|g\|_{L^1} \|\check{\varphi}\|_{L^p}$$

□

Osservazione 3.1.7

The best constant for

$$\left\| \frac{1}{|x|^\alpha} * u \right\|_{L^q} \leq C \|u\|_{L^p}$$

was found by Lieb. By duality we can get the best constant for

$$\|u\|_{L^{\frac{2d}{d-2s}}} \leq \tilde{C} \| |D|^s u \|_{L^2}.$$

⁶remember $\|f(\varepsilon x)\|_{L^p} = \varepsilon^{-\frac{d}{p}} \|f\|_{L^p}$

3.2 The Hilbert and Riesz transform

In the previous section we started talking about Rietz/Hilbert transform and how they will link the operator $|D|$ with ∇ . Let's better see this aspect. Thanks to the Sobolev inequality, when $s = 1$, we have that

$$\|u\|_{L^q} \preccurlyeq \| |D|u \|_{L^p}.$$

If $p = 2$ we saw that $\| |D|u \| = \|\nabla u\|$, unfortunately this is not the case for any p . Nevertheless we will see that for $1 < p < +\infty$ we have:

$$\| |D|u \|_{L^p} \preccurlyeq \|\nabla u\|_{L^p}.$$

In order to arrive to such an inequality we first try to write⁷ $|D|u$ in terms of ∇u .

Idea Remembering how the Fourier transform behaves for differentiation we see that:

$$2\pi|\xi| = \frac{2\pi|\xi|^2}{|\xi|} = \sum_{i=1}^d \frac{2\pi\xi_i^2}{|\xi|} = \sum_{j=1}^d (2\pi i\xi_j) \left(\frac{-i\xi_j}{|\xi|} \right).$$

Since $2\pi i\xi_j = \hat{\partial}_{x_j}$ we would love to define an operator T_j such that

$$\widehat{T_j\varphi} = -i \frac{\xi_j}{|\xi|} \hat{\varphi} \Rightarrow T_j\varphi = \left(-i \frac{\xi_j}{|\xi|} \hat{\varphi} \right)^\vee$$

then it will be clear that

$$T_j(\partial_{x_k}\varphi) = \left(-i \frac{\xi_j}{|\xi|} 2\pi i\xi_k \hat{\varphi} \right)^\vee = \partial_{x_k}(T_j\varphi).$$

So we have that the operations do commute and we have a multiplier for the operator T_j . ┘

Since the multipliers coincide seen through Fourier we have that

$$|D|\varphi = \sum_{j=1}^d \partial_{x_j}(T_j\varphi) = \sum_{j=1}^d T_j(\partial_{x_j}\varphi).$$

This means that the L^p norm can be estimated⁸ as follows:

$$\begin{aligned} \| |D|\varphi \|_{L^p} &= \left\| \sum_{j=1}^d T_j(\partial_{x_j}\varphi) \right\|_{L^p} \\ &\leq \sum_{j=1}^d \|T_j(\partial_{x_j}\varphi)\|_{L^p} \\ &\stackrel{3.3}{\preccurlyeq} \sum_{j=1}^d \|\partial_{x_j}\varphi\|_{L^p}. \end{aligned}$$

Thus we have an equivalent norm to $\|\nabla\varphi\|_{L^p}$.

⁷We have that $|D|$ is a *global operator*, whereas ∇ it's local. One can talk about *pseudodifferential operator*.

⁸The Calderón-Zygmund theorem (3.3) has a central role.

Idea The right operator T_j will be the *Riesz transform* R_j acting on Schwartz's functions. Since

$$R_j \varphi = \left(-i \frac{\xi_j}{|\xi|} \hat{\varphi} \right)^\vee = \left(-i \frac{\xi_j}{|\xi|} \right)^\vee * \varphi$$

We will need to give a sense, with the language of tempered distributions, to

$$\left(-i \frac{\xi_j}{|\xi|} \right)^\vee.$$

┘

Another reason why the Riesz transform is so important is given by the study of PDEs. When we consider the problem $-\Delta u = f$ we have that $\frac{\partial^2 u}{\partial x^2}$ have the same regularity as f . Can we say something about the mixed partials? We will have that

$$\frac{\partial^2}{\partial x_i \partial x_j} = R_i(R_j(\Delta))$$

hence the regularity of the homogeneous derivatives gives the regularity of the mixed ones if the operator R_j is continuous, i.e.:

$$\|R_j(\Delta u)\|_{L^p} \preccurlyeq \|\Delta u\|_{L^p}.$$

This will be a consequence of the *Schauder's estimates*.

3.2.1 Hilbert Transform

Let us start from the case $d = 1$. This means that $\frac{-i\xi}{|\xi|} = -i \operatorname{sign}(\xi)$.

Definition 3.2.1 – Hilbert transform

Given $\varphi \in \mathcal{S}$ we define the *Hilbert transform* as the tempered distribution such that

$$\widehat{H\varphi} = -i \operatorname{sign}(\xi) \hat{\varphi}.$$

Idea Remembering the good properties of the Fourier transform related to the convolution operator we want to find a kernel $\smile$ such that $\smile = -i \operatorname{sign}(\xi)$ then

$$H\varphi = \smile * \varphi.$$

┘

Definition 3.2.2 – Principal value

We consider $\omega_0 \in \mathcal{S}'$ such that

$$\langle \omega_0, \varphi \rangle = \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0} \int_{\{|\varepsilon| \leq |x| \leq 1\}} \frac{\varphi(x)}{x} dx + \frac{1}{\pi} \int_{\{|x| > 1\}} \frac{\varphi(x)}{x} dx$$

This is also called the *principal value* of φ .

Osservazione 3.2.1

With the previous definition it is not clear why $\omega_0 \in \mathcal{S}'$. We can rewrite $\langle \omega_0, \varphi \rangle$ as follows:

$$\langle PV(1), \varphi \rangle = \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0} \int_{|x| \geq \varepsilon} \frac{\varphi}{x} dx.$$

This second form gives more an idea of why $\omega_0 \in \mathcal{S}'$.

Proof.

Idea We will start to use a very simple, but powerful trick: if we have an odd function f on a even interval I then

$$\int_I f(x)dx = 0.$$

Dealing with computations where integrals appear we can add the quantity we need to proceed, for example to use Lagrange theorem, without any problem. $\lrcorner$

We have that $\omega_0 \in \mathcal{S}'$ if $\langle \omega_0, \varphi \rangle$ is controlled by a finite number of seminorms. This follows because

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \int_{|x| \geq \varepsilon} \frac{\varphi}{x} dx &= \int_{|\varepsilon| \leq |x| \leq 1} \frac{\varphi(x)}{x} dx + \int_{|x| > 1} \frac{x\varphi}{x^2} dx \\ &= \int_{|\varepsilon| \leq |x| \leq 1} \frac{\varphi(x) - \varphi(0)}{x} dx + \int_{|x| > 1} \frac{x\varphi}{x^2} dx \\ &\leq 2\|\varphi'\|_{L^\infty} + 2\|x\varphi\|_{L^\infty} \end{aligned}$$

where we used the Lagrange Theorem for the first term and the estimate

$$\begin{aligned} \left| \int_{|x| > 1} \frac{x\varphi}{x^2} dx \right| &\leq \int_{|x| > 1} \left| \frac{x\varphi}{x^2} \right| dx \\ &\leq \|x\varphi\|_{L^\infty} \int_{|x| > 1} \frac{1}{|x|^2} dx \\ &= 2\|x\varphi\|_{L^\infty} \end{aligned}$$

for the second one. $\square$

Definition 3.2.3 – Truncated Hilbert Transform

We consider the *truncated Hilbert transform* as

$$H^\varepsilon \varphi = \frac{1}{\pi} \int_{|y| \geq \varepsilon} \frac{f(x-y)}{y} dy = \frac{1}{\pi} \int_{|x-y| \geq \varepsilon} \frac{f(y)}{x-y} dy.$$

Osservazione 3.2.2

It is clear that we can rewrite the Hilbert's transform as

$$H\varphi = \lim_{\varepsilon \rightarrow 0} H^\varepsilon \varphi = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \int_{|x-y| \geq \varepsilon} \frac{f(y)}{x-y} dy.$$

This form is handier because it is in terms of a *singular integral*.

Idea Achtung! In this case the denominator is elevated to the power of 1 which is also the dimension in which we are working. This works because we do **not** have modules. Having an odd function we are able to prove a (p, p) -estimate. In the case of higher dimension we will have modules, hence the denominator will be of a different power, which will be estimated thanks to (HLS). $\lrcorner$

Proposition 3.2.4

The multiplier ω_0 is such that

$$\hat{\omega}_0 = -i \operatorname{sign} \xi.$$

This clearly gives us $H\varphi = \omega_0 * \varphi$.

Proof.

$$\begin{aligned} \langle \widehat{\omega_0}, \varphi \rangle &= \langle \omega_0, \hat{\varphi} \rangle \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \int_{|\xi| \geq \varepsilon} \frac{\hat{\varphi}(\xi)}{\xi} \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \int_{|\xi| \geq \varepsilon} \frac{1}{\xi} \int_{\mathbb{R}} \varphi(x) e^{-2\pi i \xi x} dx d\xi \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \int_{\mathbb{R}} \varphi(x) \int_{|\xi| \geq \varepsilon} \frac{1}{\xi} e^{-2\pi i \xi x} d\xi dx \\ &= \lim_{\varepsilon \rightarrow 0} -i \int_{\mathbb{R}} \varphi(x) \int_{|\xi| \geq \varepsilon} \frac{\sin 2\pi \xi x}{\xi} d\xi dx \\ &= \int_{\mathbb{R}} \varphi(-i \operatorname{sign}(x)) dx \\ &= \langle \varphi, -i \operatorname{sign}(x) \rangle \end{aligned}$$

where we used that $\frac{\cos(2\pi \xi x)}{\xi}$ is an odd function and the following identity:

$$\int_{-\infty}^{\infty} \frac{\sin(bx)}{x} dx = \pi \operatorname{sign}(b).$$

This follows because:

1. we have that

$$\int_{-\infty}^{\infty} \frac{\sin(x)}{x} dx = \pi.$$

Indeed if we consider $I(a) = \int_0^{+\infty} \frac{\sin(x)}{x} e^{-ax} dx$, this is equal to prove $I(0) = \frac{\pi}{2}$.

$$\begin{aligned} I'(a) &= \int_0^{+\infty} \frac{\sin(x)}{x} e^{-ax} (-x) = - \int_0^{+\infty} \sin(x) e^{-ax} dx \\ &= \left[\frac{e^{-ax}}{-a} \sin(x) \right]_0^{+\infty} + \frac{1}{a} \int_0^{+\infty} e^{-ax} \cos(x) dx \\ &= -\frac{1}{a} \left[\frac{e^{-ax}}{x} \cos(x) \right]_0^{+\infty} - \frac{1}{a} \int_0^{+\infty} e^{-ax} \sin(x) dx \\ &= -\frac{1}{a^2} - \frac{1}{a^2} I'(a). \end{aligned}$$

This gives the following ODE:

$$(1 + \frac{1}{a^2}) I' = -\frac{1}{a^2} \Rightarrow -I' = -\frac{1}{1 + a^2}$$

which solution is

$$I(a) = -\arctan(a) + C.$$

We see that sending $a \rightarrow +\infty$ we have $I(a) \rightarrow 0$ and $\arctan(a) \rightarrow \frac{\pi}{2}$, hence $C = \frac{\pi}{2}$.

2. We just need the following change of variables: $bx = x'$, indeed:

$$\begin{aligned} \int_{-\infty}^{+\infty} \frac{\sin(x')}{\frac{x'}{b}} \frac{1}{b} dx' &= \left[\int_{-\infty}^{+\infty} \frac{\sin(x)}{x} dx \right] \text{sign}(b) \\ &= \pi \text{sign}(b). \end{aligned}$$

□

We now show a really useful lemma. Define S_ω as

$$\widehat{S_\omega f} = \chi_{[-\omega, \omega]} \hat{f}$$

then

Lemma 3.2.5

If^a $\sup_{\omega>0} \|S_\omega\|_{L^p \rightarrow L^p} < +\infty$ we have

$$\|S_\omega f\|_{L^p} \preccurlyeq c_\omega \|f\|_{L^p}.$$

As a direct consequence we have that

$$\|S_\omega(f) - f\|_{L^p} \xrightarrow{\omega \rightarrow +\infty} 0.$$

^aBy $\|S_\omega\|$ we consider the operator norm of $S_\omega : L^p \rightarrow L^p$, which is $\sup_{\substack{f \in L^p \\ \|f\|_{L^p}=1}} \|S_\omega f\|_{L^p}$.

Proof of Lemma. We consider

$$A = \{f \in L^1 \cap L^p : \hat{f} \text{ has compact support}\}$$

We have that A is dense in L^p , indeed and that for any function in A we have $S_\omega(f) = f$ for $\omega \gg 1$.

$$B = \{f \in L^p : \lim_{\omega \rightarrow +\infty} S_\omega(f) = f \text{ in } L^p\}$$

is closed in L^p . (same spirit of the maximal operator proof)

We want to prove that $f_n \in B$ and $f_n \rightarrow f$ in L^p then $f \in B$. Let $\|f_n - f\| \leq \varepsilon$. We want to prove $\forall \varepsilon$:

$$\limsup_{\omega} \|S_\omega(f) - f\| \leq \varepsilon$$

By adding and subtracting the same terms we have

$$\begin{aligned} \|S_\omega(f) - f\| &= \|S_\omega(f) - S_\omega(f_n) + S_\omega(f_n) - f_n + f_n - f\| \\ &\leq \|S_\omega(f) - S_\omega(f_n)\| + \|S_\omega(f_n) - f_n\| + \|f_n - f\| \\ &= \|S_\omega(f - f_n)\| + \|S_\omega(f_n) - f_n\| + \|f_n - f\| \\ &\leq C\|f - f_n\|_{L^p} + 0 + \varepsilon \\ &\leq (1 + C)\varepsilon. \end{aligned}$$

□

We will see

$$\|Hf\|_{L^p} \preccurlyeq \|f\|_{L^p}$$

for $1 < p < \infty$. This will let us say that

Corollary 3.2.6

Is it true that $S_\omega \xrightarrow{\omega \rightarrow \infty} f$ in L^p ?

Proof of corollary. For $p = 2$ it is trivial thanks to Placherel inequality, for any p it's not so immediate. It will be true for $1 < p < \infty$. We write

$$S_\omega(f) = \int_{|\xi| < \omega} \hat{f}(\xi) e^{2\pi i \xi x} d\xi.$$

Let us see that $\sup \|S_\omega\|_{L^p \rightarrow L^p} < +\infty$ then we conclude thanks to the previous lemma.

We introduce the *Riesz Projector*

$$\begin{aligned} P : L^p &\longrightarrow L^p \\ f &\longmapsto \frac{1}{2} [f + iHf] \end{aligned}$$

We see

$$\widehat{Pf} = \frac{1}{2} [\hat{f} + \text{sign}(\xi)\hat{f}] = \chi_{[0,+\infty)}(\xi)\hat{f}.$$

Idea We want to split the interval $[-\omega, \omega]$ into two parts $[-\omega, \infty] - [\omega, \infty]$. We now, thanks to the Hilbert transform that $\widehat{Pf} = \chi_{[0,+\infty)}(\xi)\hat{f}$ is (p, p) continue. We need to prove that even its translations in Fourier stay continue. $\lrcorner$

Claim:

$$S_\omega(f) = e^{-2\pi i \omega x} P(e^{2\pi i \omega x} f) - e^{2\pi i \omega x} P(e^{-2\pi i \omega x} f)$$

which proof will be left to the reader.

We then see that

$$\|S_\omega(f)\|_{L^p} \leq \|P(e^{2\pi i \omega x} f)\|_{L^p} + \|P(e^{-2\pi i \omega x} f)\|_{L^p} \leq C\|e^{2\pi i \omega x} f\|_{L^p} + C\|e^{-2\pi i \omega x} f\|_{L^p}$$

and the exponential vanish due to the absolute value. We stress that this result is true thanks to the continuity of the Hilbert Transform. In more dimension it will not be that easy. $\square$

3.2.2 Riesz Transform

We want now to extend the Hilbert transform in the n -dimensional case. It turns out that there exists n operators in $\mathbb{R}^n$ with properties similar to H .

Definition 3.2.7 – Riesz transform

We define with $\varphi \in \mathcal{S}$ (where with j we indicate the j -th direction)

$$\langle \omega_j, \varphi \rangle = \frac{\Gamma(\frac{d+1}{2})}{\pi^{\frac{d+1}{2}}} \lim_{\varepsilon \rightarrow 0} \int_{|y| > \varepsilon} \frac{y_j}{|y|^{d+1}} \varphi(y) dy$$

and the Riesz transform

$$R_j \varphi = \frac{\Gamma(\frac{d+1}{2})}{\pi^{\frac{d+1}{2}}} \lim_{\varepsilon \rightarrow 0} \int_{|x-y| > \varepsilon} \frac{x_j - y_j}{|x-y|^{d+1}} \varphi(y) dy$$

Proposition 3.2.8

It holds

$$\widehat{R_j \varphi} = \frac{-i \xi_j}{|\xi|} \hat{\varphi}(\xi).$$

Proof.

$$\begin{aligned} \langle \hat{\omega}_j, \varphi \rangle &= \langle \omega_j, \hat{\varphi} \rangle \\ &= \frac{\Gamma(\frac{d+1}{2})}{\pi^{\frac{d+1}{2}}} \lim_{\varepsilon \rightarrow 0} \int_{|\xi| > \varepsilon} \int_{\mathbb{R}^d} \frac{\xi_j}{|\xi|} \varphi(x) e^{-2\pi i x \cdot \xi} \\ &= \frac{\Gamma(\frac{d+1}{2})}{\pi^{\frac{d+1}{2}}} \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^d} \varphi(x) \underbrace{\int_{\frac{1}{\varepsilon} > |\xi| > \varepsilon} \frac{\xi_j}{|\xi|} e^{-2\pi i x \cdot \xi}}_{=-i \frac{x_j}{|x|}} \end{aligned}$$

Rewriting ξ as $r\theta$ with $r > 0$ e $\theta \in \mathbb{S}^{d-1}$ we can carry on the last estimate as:

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \int_{\frac{1}{\varepsilon} > |\xi| > \varepsilon} \frac{\xi_j}{|\xi|} e^{-2\pi i x \cdot \xi} &= \lim \int_{\mathbb{S}^{d-1}} \int_{\frac{1}{\varepsilon} > r > \varepsilon} \frac{r \theta_j e^{-2\pi i r \theta \cdot x}}{r^{d+1}} r^{d-1} dr d\sigma \\ &= \lim \int_{\mathbb{S}^{d-1}} \int_{\frac{1}{\varepsilon} > r > \varepsilon} \frac{\theta_j e^{-2\pi i r \theta \cdot x}}{r} dr d\sigma \\ &= -i \lim \int_{\mathbb{S}^{d-1}} \int_{\frac{1}{\varepsilon} > r > \varepsilon} \frac{\theta_j \sin(2\pi r \theta \cdot x)}{r} dr d\sigma \\ &= -\frac{i\pi}{2} \int_{\mathbb{S}^{d-1}} \theta_j \operatorname{sign}(\theta \cdot x) d\sigma \\ &= \frac{i\pi}{2} \left(\frac{\Gamma(\frac{d+1}{2})}{\pi^{d+1}} \right)^{-1}. \end{aligned}$$

where the last equality will be proved in the next proposition. $\square$

Proposition 3.2.9

It holds that

$$\int_{S^{d-1}} \text{sign}(\theta \cdot \xi) \theta_j d\theta = \frac{(2\pi)^{\frac{d-1}{2}} \xi_j}{\Gamma(\frac{d-1}{2}) |\xi|}$$

Proof. We start by recalling that, by using spherical coordinates

$$\int_{RS^{d-1}} f(x) d\sigma = \int_0^\pi \dots \int_0^\pi \int_0^{2\pi} f(x(\varphi)) J(d, R, \varphi) d\varphi_1 \dots d\varphi_{d-1}$$

where

$$J = R^{d-1} (\sin \varphi_1)^{d-2} \dots (\sin \varphi_{d-3})^2 (\sin \varphi_{d-2}).$$

The next step is proving that

$$\int_{RS^{d-1}} f(x) d\sigma = \int_{-R}^R \int_{\sqrt{R^2-s^2}}^R f(s, \theta) d\theta \frac{R ds}{\sqrt{R^2-s^2}}. \quad (3.1)$$

In fact, taking $\varphi = (\varphi_2, \dots, \varphi_{d-1})$ and $x' = x'(\varphi') = (\cos \varphi_2, \sin \varphi_2 \sin \varphi_3, \dots, \sin \varphi_2 \dots \sin \varphi_{d-1})$ we have that

$$\begin{aligned} \int_{RS^{d-1}} f(x) d\sigma &= \int_0^\pi \left[\int_{S^{d-2}} f(R \cos \varphi_1, R \sin \varphi_1 x') d\sigma \right] R^{d-1} (\sin \varphi_1)^{d-2} d\varphi_1 \\ &= \int_{-R}^R \int_{S^{d-2}} f(s, \sqrt{R^2-s^2} x') d\sigma R^{d-1} (\sin \varphi_1)^{d-2} \frac{ds}{\sqrt{R^2-s^2}} \\ &= \int_{-R}^R \int_{S^{d-2}} f(s, \sqrt{R^2-s^2} x') d\sigma R^{d-1} (\sqrt{R^2-s^2})^{d-3} ds \\ &= \int_{-R}^R \int f(s, \theta) d\theta \frac{R}{\sqrt{R^2-s^2}} ds \end{aligned}$$

where we used the change of variable $s = R \cos \varphi_1$. Now we can proceed with the proof. Choose the orthogonal matrix A such that $Ae_j = \xi$, then

$$\begin{aligned} \int_{S^{d-1}} \text{sign}(\xi \cdot \theta) \theta_j d\theta &= \int_{S^{d-1}} \text{sign}(Ae_j \cdot \theta) \theta_j d\theta \\ &= \int_{S^{d-1}} \text{sign}(e_j \cdot A^T \theta) (AA^T \theta)_j d\theta \\ &= \int_{S^{d-1}} \text{sign}(e_j \cdot \tilde{\theta}) (A\tilde{\theta})_j d\tilde{\theta} \\ &= \int_{S^{d-1}} \text{sign}(\tilde{\theta}_j) (A\tilde{\theta})_j d\tilde{\theta} \\ &= \xi_j \int_{S^{d-1}} |\tilde{\theta}_j| d\tilde{\theta} \end{aligned}$$

where we used the change of variable $\tilde{\theta} = A^T \theta$ and we noticed that, since

$$(A\tilde{\theta})_j = a_{j1} \tilde{\theta}_1 + \dots + \xi_j \tilde{\theta}_j + \dots + a_{jd} \tilde{\theta}_d$$

and the integral

$$\int_{S^{d-1}} \text{sign}(\tilde{\theta}_j) \tilde{\theta}_k d\tilde{\theta} = 0 \quad \text{if } j \neq k$$

just one term survives. At this point we can apply (3.1) and conclude

$$\begin{aligned}\xi_j \int_{S^{d-1}} |\tilde{\theta}_j| d\tilde{\theta} &= \xi_j \int_{-1}^1 \int_{S^{d-2}} d\tilde{\theta}' |s| (1-s^2)^{\frac{d-3}{2}} ds \\ &= \xi_j \omega_{d-2} \int_0^1 \int_{S^{d-2}} u^{\frac{d-3}{2}} du \\ &= \frac{2\pi^{\frac{d-3}{2}}}{\Gamma(\frac{d+1}{2})} \xi_j.\end{aligned}$$

□

3.3 Calderón-Zygmund's theory

Definition 3.3.1 – Calderón-Zygmund kernel

We define the *Calderón-Zygmund kernel* as a function $K : \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{C}$ such that:

1. $|K(x)| \leq \beta |x|^{-d}$ for every $x \neq 0$.
2. *Hörmander's condition*: there exists B such that $\forall y \neq 0$

$$\int_{|x| \geq 2|y|} |K(x) - K(x-y)| dx \leq B$$

3. *Zero-Average on shells*: We have $\forall R, S > 0$:

$$\int_{R < |x| < S} K(x) dx = 0.$$

Osservazione 3.3.1

These three condition will not always be used. We will specify when this is the case or not.

Definition 3.3.2 – Singular integrals

Given a Calderón-Zygmund kernel, we define an operator by means of its principal value. The *singular integral operator* with kernel K is defined as follows:

$$Tf := \lim_{\varepsilon \rightarrow 0} \int_{|x-y| \geq \varepsilon} K(x-y) f(y) dy.$$

Osservazione 3.3.2

If K is such that $|\nabla K| \lesssim \frac{1}{|x|^{d+1}}$ for all $x \neq 0$, then the Hörmander's condition holds.

Proof. We define $g(t) = K(x - ty)$, then

$$\begin{aligned}K(x) - K(x-y) &= g(0) - g(1) \\ &= \int_0^1 g'(s) ds \\ &= - \int_0^1 \nabla K(x - sy) \cdot y ds\end{aligned}$$

At this point

$$\begin{aligned}
\int_{|x|>2|y|} |K(x) - K(x-y)| &= \int_{|x|>2|y|} \left| \int_0^1 \nabla K(x-sy) \cdot y ds \right| dx \\
&\leq \int_{|x|>2|y|} \int_0^1 |\nabla K(x-sy) \cdot y| ds dx \\
&\leq \int_{|x|>2|y|} \|y\| \int_0^1 |\nabla K(x-sy)| ds dx \\
&\leq C \int_{|x|>2|y|} \|y\| \int_0^1 \frac{ds}{|x-sy|^{d+1}} dx
\end{aligned}$$

Since $|x-sy| > |y| - s|y|$ and $s|y| < |y| < \frac{1}{2}|x|$ we have $|x-sy| > \frac{1}{2}|x|$. The last integral can then be estimated as

$$\begin{aligned}
C \int_{|x|>2|y|} \|y\| \int_0^1 \frac{ds}{|x-sy|^{d+1}} dx &\leq \tilde{C} \int_{|x|>2|y|} \|y\| \int_0^1 \frac{ds}{|x|^{d+1}} dx \\
&\leq C|y|\omega_{d-1} \int_{2|y|}^{\infty} \frac{1}{r^{d+1}} r^{d-1} dr \\
&\simeq |y|\omega_{d-1} \left(-\frac{1}{t} \Big|_{2|y|}^{\infty} \right) \\
&\simeq \omega_{d-1}
\end{aligned}$$

□

Theorem 3.3.3 – Calderón-Zygmund's Theorem

We have

$$\|Tf\|_{L^p} \lesssim \|f\|_{L^p}.$$

Idea The proof of this theorem will follow the next steps:

1. We prove the $L^2 - L^2$ strong continuity.
2. We prove the $L^1 - L^1$ weak continuity.
3. By using the Marcinkiewicz theorem we have $L^p - L^p$ strong continuity for $1 < p \leq 2$.
4. By duality we get $L^p - L^p$ strong continuity for $2 \leq p < +\infty$.

┘

Definition 3.3.4

For any $r, s \in \mathbb{R}^+$ we can consider the *truncation kernel* $K_{r,s}$ defined as:

$$K_{r,s}(x) = K(x)\chi_{r<|x|<s}(x).$$

This truncation will be useful in order to avoid singularity at 0 and at infinity.

Lemma 3.3.5 – Strong $L^2 - L^2$ continuity

Given a Calderón-Zygmund kernel K it holds that

$$\|K_{r,s} * f\|_{L^2} \preccurlyeq \|f\|_{L^2} \text{ uniformly in } r, s.$$

Remembering that $K * f = \lim_{\substack{r \rightarrow 0 \\ s \rightarrow +\infty}} K_{r,s} * f$, then we have:

$$\|K * f\|_{L^2} \leq C \|f\|_{L^2}.$$

Proof. In this proof we suppose $f \in \mathcal{S}$. If we consider

$$K_\varepsilon * f = \int_{\varepsilon < |y| < \frac{1}{\varepsilon}} K(y) f(x-y) dy$$

the $L^2 - L^2$ estimate will follow from the uniform convergence and the fact that $\{K_\varepsilon * f\}$ is a Cauchy sequence in L^2 .

$\{K_\varepsilon * f\}$ is a Cauchy sequence: given $\varepsilon_2 < \varepsilon_1$ we want to study

$$\begin{aligned} \|K_{\varepsilon_2} * f - K_{\varepsilon_1} * f\|_{L_x^2} &= \left\| \int_{\varepsilon_2 < |y| < \frac{1}{\varepsilon_2}} K(y) f(x-y) dx - \int_{\varepsilon_1 < |y| < \frac{1}{\varepsilon_1}} K(y) f(x-y) dx \right\|_{L_x^2} \\ &= \left\| \int_{\varepsilon_2 < |y| < \varepsilon_1} K(y) f(x-y) dx - \int_{\frac{1}{\varepsilon_1} < |y| < \frac{1}{\varepsilon_2}} K(y) f(x-y) dx \right\|_{L_x^2}. \end{aligned}$$

We now compute the two integrals separately. We first note that:

$$\begin{aligned} \left\| K \chi_{\frac{1}{\varepsilon_1} < |y| < \frac{1}{\varepsilon_2}} \right\|_{L^2} &= \left(\int_{\frac{1}{\varepsilon_1} < |y| < \frac{1}{\varepsilon_2}} |K(x)|^2 dx \right)^{\frac{1}{2}} \\ &\preccurlyeq \left(\int \frac{1}{|x|^{2d}} \right)^{\frac{1}{2}} \\ &= \left(\omega_{d-1} \int \frac{r^{d-1}}{r^{2d}} \right)^{\frac{1}{2}} \approx \varepsilon_1^{\frac{d}{2}}. \end{aligned}$$

By Young's interpolation theorem we have

$$\begin{aligned} \left\| \int_{\frac{1}{\varepsilon_1} < |y| < \frac{1}{\varepsilon_2}} K(y) f(x-y) dx \right\|_{L^2} &\leq \left\| K \chi_{\frac{1}{\varepsilon_1} < |y| < \frac{1}{\varepsilon_2}} \right\|_{L^2} \|f\|_{L^1} \\ &\preccurlyeq \|f\|_1 \varepsilon_1^{\frac{d}{2}}. \end{aligned}$$

For the second integral we use the third condition in the definition⁹ of the Calderón-Zygmund kernel, i.e.

$$\int_{\varepsilon_2 < |y| < \varepsilon_1} K(y) f(x-y) dx = \int_{\varepsilon_2 < |y| < \varepsilon_1} K(y) (f(x-y) - f(x)) dx.$$

⁹the trick is the following: having and integral equal to 0 we can add the quantity we need in order to proceed in the computation without changing it.

Having $f \in \mathcal{S}$ we can write $f(x-y) - f(x) = \int_0^1 \langle \nabla f(x-ty), y \rangle dt$ and, by Minkowski's theorem, it follows that:

$$\begin{aligned}
\left\| \int_{\varepsilon_2 < |y| < \varepsilon_1} K(y) \int_0^1 \nabla f(x-ty) \cdot y dt dy \right\| &\leq \int_0^1 \int_{\varepsilon_2 < |y| < \varepsilon_1} |y| |K(y)| \|\nabla f(x-ty)\|_{L_x^2} dy dt \\
&\preccurlyeq \|\nabla f\|_{L^2} \int_0^1 \int_{\varepsilon_2 \leq |y| \leq \varepsilon_1} \frac{1}{|y|^{d-1}} \\
&= \|\nabla f\|_{\omega_{d_1}} \int_0^1 \int_{\varepsilon_2}^{\varepsilon_1} \frac{r^{d-1}}{r^{d-1}} \\
&\preccurlyeq \varepsilon_1
\end{aligned}$$

where we used the first condition, i.e $|y| |K(y)| \leq \frac{1}{|y|^{d-1}}$ and the fact that the norm is translation invariant.

Uniform estimate: For this part of the proof we will consider the case of $r = \frac{1}{\varepsilon}$ and $s = \varepsilon$, so the kernel will be

$$K_\varepsilon = K \chi_{\frac{1}{\varepsilon} \leq |x| \leq \varepsilon}$$

One can see that, if K is a Calderón-Zygmund kernel, then also K_ε is one. If we want an uniform estimate on K_ε , thanks to Plancherel, we can check it on $\hat{K}_\varepsilon$.

$$\begin{aligned}
\hat{K}_\varepsilon &= \int_{\mathbb{R}^d} K_\varepsilon e^{-2\pi i \xi x} dx \\
&= \int_{|x| \leq \frac{1}{|\xi|}} K_\varepsilon e^{-2\pi i \xi x} dx + \int_{|x| \geq \frac{1}{|\xi|}} K_\varepsilon e^{-2\pi i \xi x} dx \\
&= I + II.
\end{aligned}$$

We now estimate the two integrals separately. We can rewrite¹⁰:

$$\begin{aligned}
|I| &= \left| \int_{\frac{1}{\varepsilon} < |x| < \frac{1}{|\xi|}} K(x) (e^{-2\pi i \xi \cdot x} - 1) dx \right| \\
&\leq \int_{\frac{1}{\varepsilon} < |x| < \frac{1}{|\xi|}} |K(x)| 2\pi |\xi| |x| dx \\
&= 2\pi |\xi| \int_{\frac{1}{\varepsilon} < |x| < \frac{1}{|\xi|}} |K(x)| |x| dx \\
&\leq 2\pi |\xi| \int_{\frac{1}{\varepsilon}}^{\frac{1}{|\xi|}} \frac{1}{|x|^{d-1}} \\
&\approx 2\pi |\xi| \frac{1}{|\xi|} \approx 1.
\end{aligned}$$

Idea The estimate of II starts with a computation that, apparently is nonsense, but that it will be handy since it allows us to use the Hörmander condition, which we still have not

¹⁰We use the trick on shell and the Lagrange estimate with $F(x) = e^{-2\pi i \xi \cdot x}$.

used. We can write

$$\begin{aligned}
 II &= \int_{\frac{1}{|\xi|} \leq |x| < \varepsilon} K(x) e^{-2\pi i \xi \cdot x} \\
 &= - \int_{\frac{1}{|\xi|} \leq |x| < \varepsilon} K(x) e^{-2\pi i \xi \cdot \left(x + \frac{\xi}{2|\xi|^2}\right)} dx \\
 &= - \underbrace{\int_{\frac{1}{|\xi|} < |x - \frac{\xi}{2|\xi|^2}| < \varepsilon} e^{-2\pi i \xi \cdot x} K\left(x - \frac{\xi}{2|\xi|^2}\right) dx}_{B}.
 \end{aligned}$$

So we have an equality like $II = -B$. We can also write $2II = II - B$. The key passage will be to add and subtract the right quantity A so that we can use Hörmander and have

$$2II = (II - A) + (-B + A).$$

One can see that the right A is:

$$A = \int_{\frac{1}{|\xi|} \leq |x| \leq \varepsilon} K\left(x - \frac{\xi}{2|\xi|^2}\right) e^{-2\pi i \xi \cdot x} dx.$$

⌋

We can estimate the first term:

$$\begin{aligned}
 |II - A| &= \left| \int_{\frac{1}{|\xi|} \leq |x| \leq \varepsilon} \left(K(x) - K\left(x - \frac{\xi}{2|\xi|^2}\right) \right) e^{-2\pi i \xi \cdot x} dx \right| \\
 &\leq \int_{\frac{1}{|\xi|} \leq |x| \leq \varepsilon} \left| K(x) - K\left(x - \frac{\xi}{2|\xi|^2}\right) \right| dx \\
 &\preccurlyeq 1
 \end{aligned}$$

where we use the Hörmander condition¹¹ on $|x| > \frac{1}{|\xi|} = 2|y|$.

For the other term we can see instead that:

$$A - B = \int_{\frac{1}{|\xi|} < |x|} K_\varepsilon\left(x - \frac{\xi}{2|\xi|^2}\right) e^{-2\pi i \xi \cdot x} dx - \int_{\frac{1}{|\xi|} < \left|x - \frac{\xi}{2|\xi|^2}\right|} K_\varepsilon\left(x - \frac{\xi}{2|\xi|^2}\right) e^{-2\pi i \xi \cdot x} dx$$

If we explicit the inequality in the domain of the integral we have¹²:

$$\left| x - \frac{\xi}{2|\xi|^2} \right| \geq |x| - \left| \frac{\xi}{2|\xi|^2} \right| = |x| - \frac{1}{2|\xi|} \geq \frac{1}{2|\xi|}.$$

$$\begin{aligned}
 \left| \int_{\left|x - \frac{\xi}{2|\xi|^2}\right| \leq |\xi| \leq |x| \leq s} K\left(x - \frac{\xi}{2|\xi|^2}\right) e^{-2\pi i \xi \cdot x} dx \right| &\leq \int_{\frac{1}{2|\xi|} \leq \left|x - \frac{\xi}{2|\xi|^2}\right| \leq \frac{1}{|\xi|}} |K\left(x - \frac{\xi}{2|\xi|^2}\right) e^{-2\pi i \xi \cdot x}| dx \\
 &= \int_{\frac{1}{2|\xi|} \leq |x| \leq \frac{1}{|\xi|}} |K(x)| dx \\
 &\leq \int_{\frac{1}{2|\xi|}}^{\frac{1}{|\xi|}} \frac{r^{d-1}}{r^d} dr = \log(2)
 \end{aligned}$$

¹¹The change of variables is: $\frac{\xi}{2|\xi|^2} = y$. This gives $|y| = \frac{1}{2|\xi|}$.

¹²since $\frac{1}{|\xi|} < |x|$ then $-\frac{1}{2|\xi|} > -\frac{|x|}{2}$.

which does not depend on ξ .

Conclusion: Using Plancherel we can see that

$$\|K_\varepsilon * f\|_{L^2} = \|\widehat{K_\varepsilon f}\|_{L^2} \leq \sup_{\varepsilon > 0} \|\widehat{K_\varepsilon}\|_{L^2} \|\widehat{f}\|_{L^2} \leq \sup_{\varepsilon > 0} \|\widehat{K_\varepsilon}\|_{L^2} \|f\|_{L^2} \preccurlyeq \|f\|_{L^2}$$

Remembering that, in the sense of the principal valute, we have

$$Tf = \int K(x-y)f(y)dy$$

then we have the $L^2 - L^2$ estimate:

$$\|Tf\|_{L^2} \preccurlyeq \|f\|_{L^2} \quad \forall f \in \mathcal{S}.$$

By a density argument this follows also $\forall f \in L^p$. □

Osservazione 3.3.3

For the Hilbert and the Riesz transform this estimate is obvious. It follows directly from the Plancherel formula.

Lemma 3.3.6 – Weak $L^1 - L^1$ continuity

Given the kernel K and the estimate $\|Tf\|_{L^2} \preccurlyeq \|f\|_{L^2}$ we have that:

$$\mu\{|K * f| > \lambda\} \preccurlyeq \frac{\|f\|_{L^1}}{\lambda}.$$

Osservazione 3.3.4

This lemma actually holds if we know that $\|Tf\|_{L^2} \preccurlyeq \|f\|_{L^2}$ and if we have a kernel M which satisfies the Hörmander condition. In our case the $L^2 - L^2$ estimate is guaranteed thanks to the Calderón-Zygmund kernel.

In order to prove this lemma first we need to see the following:

Lemma 3.3.7 – Calderón-Zygmund Decomposition Lemma

If we have $f \in L^1(\mathbb{R}^d)$ and $\lambda > 0$ we can write^a $f = g + b$, with this two properties:

$$\begin{cases} |g| \leq \lambda \\ b = \sum_{Q \in \mathcal{Q}} \chi_Q f \end{cases}$$

where $\mathcal{Q}$ is a family of disjoint cubes, with the property:

$$\lambda \leq \frac{1}{|Q|} \int_Q |f| dx \leq 2^d \lambda.$$

Furthermore,

$$\left| \bigcup_{Q \in \mathcal{Q}} Q \right| \leq \frac{1}{\lambda} \|f\|_1.$$

^a g denotes good function and b bad function.

Proof. For all $\ell \in \mathbb{Z}$ we define D^ℓ a family of dyadic cubes:

$$D^\ell = \left\{ \prod_{i=1}^d \left[2^\ell m_i, 2^\ell(m_i + 1) \right) \mid m_1, \dots, m_d \in \mathbb{Z} \right\}.$$

If the cube $Q \in D^\ell$ and $Q' \in D^{\ell'}$ there are two possibilities:

1. $Q \cap Q' = \emptyset$
2. $Q \subset Q'$ or $Q' \subset Q$.

Fix λ then for $f \in L^1$ we have that there exists $\ell_0, \forall Q \in D^{\ell_0}$ such that¹³

$$\frac{1}{|Q|} \int_Q |f| dx \leq \lambda.$$

For any cube in D^{ℓ_0} we take the¹⁴ "children" Q' , i.e. $Q' \in D^{\ell_0-1}$. With the children it may happen that

$$\frac{1}{|Q'|} \int_{Q'} |f| dx < \lambda \quad \text{or} \quad \frac{1}{|Q'|} \int_{Q'} |f| dx \geq \lambda.$$

If the first condition happens, we say Q' is a *good set* and iterate the construction, otherwise Q' is a *bad set*. We define $\mathcal{B}$ as the family of bad sets. We now ask ourselves: if $Q \in \mathcal{B}$ do we have an upper bound for the integral?

$$\lambda \leq \frac{1}{|Q'|} \int_{Q'} |f| dx = \frac{2^d}{|Q|} \int_{Q'} |f| dx \leq 2^d \lambda.$$

Idea If we take a point which $\notin \mathcal{B}$, it means that each cube Q which surrounds it, has integral $< \lambda$. The thesis will follow thanks to Lebesgue theorem. $\lrcorner$

What about the measure of the bad family?

$$\left| \bigcup_{Q \in \mathcal{B}} Q \right| \leq \sum_{Q \in \mathcal{B}} |Q| \leq \sum \frac{1}{\lambda} \int_Q |f| dx \leq \frac{1}{\lambda} \|f\|_{L^1}.$$

Let us take $x_0 \in \mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} \overline{Q}$. It means that there exists Q_i such that $|Q_i| \rightarrow 0$, $x_0 \in Q_i$ and

$$\frac{1}{|Q_i|} \int_{Q_i} |f| dx \leq \lambda$$

Then for lebesgue theorem we have $|f(x_0)| \leq \lambda$ a.e. We can define the good function $g := f - \sum_Q \chi_{\overline{Q}} f$ and have the thesis. $\square$

Proof of Weak $L^1 - L^1$ continuity. Let us define

$$f_1 = \begin{cases} f & \mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q \\ \frac{1}{|Q|} \int_Q f dx & \text{otherwise} \end{cases} \quad f_2 = \begin{cases} 0 & \mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q \\ f - \frac{1}{|Q|} \int_Q f dx & Q \in \mathcal{B} \end{cases}$$

We have that $f = f_1 + f_2$, where

¹³Having $f \in L^1$ and that $|Q| = 2^{d\ell_0}$ we can determine ℓ_0 such that $\frac{\|f\|_{L^1}}{\lambda} < 2^{d\ell_0}$.

¹⁴It is a cube with half the side length of the previous one.

1. $\|f_1\|_{L^\infty} \leq 2^d \lambda$, indeed if $x \in \mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q$ we have that $|f| < \lambda$, otherwise $\frac{1}{|Q|} \int_Q f dx < 2^d \lambda$.
2. $\|f_1\|_{L^1} \leq \|f\|_{L^1}$, indeed

$$\int_Q \left| \frac{1}{|Q|} \int_Q f dx \right| dy \leq \int_Q |f| dx$$

3. $\|f_2\|_{L^1} \leq 2\|f\|_{L^1}$, it is similar to the previous one.

In order to prove the $(1-1)$ -weak condition we can write:

$$\mu\{|K_\varepsilon * f| > \lambda\} \leq \underbrace{\mu\left\{|K_\varepsilon * f_1| > \frac{\lambda}{2}\right\}}_I + \underbrace{\mu\left\{|K_\varepsilon * f_2| > \frac{\lambda}{2}\right\}}_{II}$$

Knowing $\int |f_1|^2 dx \leq 2^d \lambda \|f_1\|_{L^1}$, thanks to the $(2-2)$ -continuity and to the chebychev inequality we have:

$$I \leq C \frac{\|f_1\|_{L^2}^2}{\lambda^2} \preccurlyeq \frac{\|f\|_{L^1}}{\lambda}.$$

Idea In order to estimate II it will be cunning to consider bigger cubes Q^* in order to separate points but having still control. $\lrcorner$

Given $Q \in \mathcal{B}$ which side is of length ℓ , we consider Q^* such that its size is $\ell^* = 2\sqrt{d}\ell$. We can write

$$\begin{aligned} II &\leq \mu\left(\bigcup_{Q \in \mathcal{B}} Q^*\right) + \mu\left\{\mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q^*, |K_\varepsilon * f_2| > \frac{\lambda}{2}\right\} \\ &\leq \sum_{Q \in \mathcal{B}} \mu(Q^*) + \mu\left\{\mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q^*, |K_\varepsilon * f_2| > \frac{\lambda}{2}\right\} \\ &= \sum (2\sqrt{d})^d \mu(Q) + \mu\left\{\mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q^*, |K_\varepsilon * f_2| > \frac{\lambda}{2}\right\} \\ &\preccurlyeq \frac{\|f\|_{L^1}}{\lambda} + \mu\left\{\mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q^*, |K_\varepsilon * f_2| > \frac{\lambda}{2}\right\}. \end{aligned}$$

We have the thesis if we prove that

$$\underbrace{\mu\left\{\mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q^*, |K_\varepsilon * f_2| > \frac{\lambda}{2}\right\}}_C \preccurlyeq \frac{\|f\|_{L^1}}{\lambda}$$

If we consider $f_Q = (f - f_Q |f| dx) \chi_Q$, by Chebychev we have

$$C \preccurlyeq \frac{1}{\lambda} \int_{\mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{B}} Q^*} |K_\varepsilon * f_2| dx \preccurlyeq \sum_Q \frac{1}{\lambda} \int_{\mathbb{R}^d \setminus Q^*} |K_\varepsilon * f_Q| dx$$

We can find y_Q such that each of its neighbourhood is contained in Q . Then we have¹⁵:

$$\begin{aligned} K_\varepsilon * f_Q &= \int_Q K_\varepsilon(x-y) f_Q(y) dy \\ &= \int_Q [K_\varepsilon(x-y) - K_\varepsilon(x-y_Q)] f_Q(y) dy. \end{aligned}$$

¹⁵ f_Q has zero average for definition

We have proved that

$$C \preccurlyeq \frac{1}{\lambda} \sum_{\mathcal{Q}} \int_{\mathbb{R}^d \setminus \mathcal{Q}^*} \left| \int_{\mathcal{Q}} [K_\varepsilon(x-y) - K_\varepsilon(x-y_{\mathcal{Q}})] f_{\mathcal{Q}}(y) dy \right| dx.$$

We conclude thanks to the Hörmander condition and Fubini:

$$\begin{aligned} C &\preccurlyeq \frac{1}{\lambda} \sum_{\mathcal{Q}} \int_{\mathcal{Q}} \int_{\mathbb{R}^d \setminus \mathcal{Q}^*} |[K_\varepsilon(x-y) - K_\varepsilon(x-y_{\mathcal{Q}})] f_{\mathcal{Q}}(y)| dx dy \\ &\preccurlyeq \frac{1}{\lambda} \sum_{\mathcal{Q} \in \mathcal{B}} \int_{\mathcal{Q}} |f_{\mathcal{Q}}| dx \\ &\leq \frac{1}{\lambda} \|f\|_{L^1}. \end{aligned}$$

□

Proof of Theorem. For the previous lemmas we get the $L^2 - L^2$ strong continuity and the $L^1 - L^1$ weak continuity. By applying the Marcinkiewitz theorem we have the $L^p - L^p$ strong continuity for $1 < p \leq 2$. We show now the $L^p - L^p$ strong continuity for $2 \leq p < \infty$ by a *duality argument*: given T , let us consider the *adjoint operator* T^* :

$$\langle Tf, g \rangle = \langle f, T^*g \rangle.$$

So T^* is defined as

$$T^*g = \int K^*(x-y)g(y)dy$$

where

$$K^* = \overline{K(-x)}.$$

By definition of the $\|\cdot\|_{L^p}$ we have

$$\|Tf\|_{L^p} = \sup_{\|g\|_{L^{p'}}=1} |\langle Tf, g \rangle| = \sup_{\|g\|_{L^{p'}}=1} |\langle f, T^*g \rangle|.$$

So we have

$$|\langle f, T^*g \rangle| \leq \|f\|_{L^p} \|T^*g\|_{L^{p'}} \preccurlyeq \|f\|_{L^p} \|g\|_{L^{p'}}$$

and

$$\|Tf\|_{L^p} \preccurlyeq \|f\|_{L^p}.$$

□

3.4 Littlewood-Paley theory

We now present the last result seen in this course, which was proved at the beginning of the 1960s and is nowadays one of the most common tool in harmonic analysis.

Lemma 3.4.1 – Paley-Littlewood decomposition

There exists $\psi \in C^\infty(\mathbb{R}^d)$ with compact support in $\mathbb{R}^d \setminus \{0\}$ such that

$$\sum_{j=-\infty}^{\infty} \psi(2^{-j}x) = 1 \quad \forall x \neq 0$$

For every $x \neq 0$ at max two of the terms overlaps. We can also take ψ radially symmetric.

Osservazione 3.4.1

This function ψ can be seen as a *locator*, i.e. when we consider $\psi \hat{f}$ we are localizing the function to a specific frequency. Indedd we can see that

$$\hat{f} = \sum_{j=-\infty}^{\infty} \psi(2^{-j}x) \hat{f} = \sum_{j=-\infty}^{+\infty} \hat{f}_j.$$

Proof. Let us consider $\chi(x) = \begin{cases} 1 & |x| \leq 1 \\ 0 & |x| \geq 2 \end{cases}$ such that $\chi \in C^\infty$. We define $\psi(x) = \chi(x) - \chi(2x)$. This function is such that¹⁶:

$$\begin{aligned} \sum_{j=-N}^N \psi(2^{-j}x) &= \psi(2^N x) + \psi(2^{N-1}x) + \dots + \psi(2^{-N}x) \\ &= \chi(2^{-N}x) - \chi(2^{N+1}x) \end{aligned}$$

Passing to the limit $N \rightarrow +\infty$ we have the thesis since $\chi(0) - \chi(\infty) = 1$. $\square$

The following result has a proof which is a prototype use of the locator we just defined.

Theorem 3.4.2 – Mikhlin- Hörmander

Let $m : \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{C}$ such that $\forall |\gamma| \leq d+2$ it holds

$$|\partial^\gamma m(z)| \lesssim |\xi|^{-|\gamma|}.$$

Then for all $1 < p < \infty$

$$\left\| \left(m(\xi) \hat{f} \right)^\vee \right\|_{L^p} \lesssim \|f\|_{L^p}.$$

Idea We proved that having a Calderón-Zygmund kernel K gives us the estimate $\|Tf\|_{L^p} \lesssim \|f\|_{L^p}$. In this statement it is as if we were considering m such that $\hat{K} = m$, hence we are unloading the requests on the kernel as decay conditions on the multiplier. $\lrcorner$

¹⁶We have a telescopic sum.

Proof. If we consider $\gamma = 0$ then it is clear that $m \in L^\infty$ and so, for Plancherel:

$$\left\| \left(m(\xi) \hat{f} \right)^\vee \right\|_{L^2} = \|m(\xi) \hat{f}\|_{L^2} \leq \|\hat{f}\|_{L^2} = \|f\|_{L^2}.$$

This gives us a $(2, 2)$ -estimate. As we saw in Observation (3.3.4) we just need to prove that the Hörmander condition holds. Let us write¹⁷:

$$\begin{aligned} m(\xi) &= \sum_{N \in 2^{\mathbb{Z}}} \psi\left(\frac{x}{N}\right) m(\xi) \\ &= \sum_{N \in 2^{\mathbb{Z}}} \psi_N(\xi) m(\xi) \\ &= \sum_{N \in 2^{\mathbb{Z}}} m_N(\xi). \end{aligned}$$

Observation (3.3.2) guarantees us that, if $|\nabla K| \lesssim \frac{1}{|x|^{d+1}}$, then the Hörmander condition holds. But we remember that $K = \check{m}$, hence:

$$|\nabla K| \leq \sum_{N \in 2^{\mathbb{Z}}} |\nabla(\check{m}_N)|.$$

Let us estimate $\|x^\alpha \nabla \check{m}_N\|_{L^\infty}$. Remembering that $\|f\|_{L^\infty} \leq \|\hat{f}\|_{L^1}$ and that $\partial^\alpha \hat{f} = ((-2\pi i x)|x|^\alpha f)$, we have that

$$\|x^\alpha \nabla \check{m}_N\|_{L^\infty} \lesssim \|\partial^\alpha \xi m_N\|_{L^1}.$$

It holds¹⁸:

$$\begin{aligned} |\partial^\alpha(\xi m_N)| &= |\partial^\alpha(\xi \psi_N(\xi) m(\xi))| \lesssim \sum_{\alpha_1 + \alpha_2 = \alpha} |\partial^{\alpha_1}(\xi m(\xi))| |\partial^{\alpha_2}(\psi_N(\xi))| \\ &= \sum_{\alpha_1 + \alpha_2 = \alpha} |\partial^{\alpha_1}(\xi m)| \left| \partial^{\alpha_2} \psi\left(\frac{\xi}{N}\right) \right| \\ &\lesssim \sum_{\alpha_1 + \alpha_2 = \alpha} \frac{1}{|\xi|^{|\alpha_1| - 1}} N^{-|\alpha_2|} \left| \partial^{\alpha_2} \psi\left(\frac{x}{N}\right) \right| \\ &\lesssim \sum_{\alpha_1 + \alpha_2 = \alpha} |\xi|^{1 - |\alpha_1|} N^{-|\alpha_2|} \left| \partial^{\alpha_2} \psi\left(\frac{\xi}{N}\right) \right|. \end{aligned}$$

Idea The advantage of using m_N is that the function is now localized, i.e. $\int_{\mathbb{R}^d} = \int_{|\xi| \sim N}$. $\dashv$

¹⁷This is a standard abbreviation to write the dyadic numbers

¹⁸we remember that $|\partial^{\alpha_1} m| \lesssim \frac{1}{|\xi|^{|\alpha_1|}}$

Proceeding in the computation we get:

$$\begin{aligned}
\|x^\alpha \nabla \check{m}_N\|_{L^\infty} &\leq \sum_{\alpha_1+\alpha_2=\alpha} \int_{\mathbb{R}^d} |\xi|^{1-|\alpha_1|} N^{-|\alpha_2|} |\partial^{\alpha_2} \psi\left(\frac{\xi}{N}\right)| d\xi \\
&\leq \sum_{\alpha_1+\alpha_2=\alpha} \int_{|\xi| \sim N} |\xi|^{1-|\alpha_1|} N^{-|\alpha_2|} |\partial^{\alpha_2} \psi\left(\frac{\xi}{N}\right)| d\xi \\
&= \sum_{\alpha_1+\alpha_2=\alpha} \int_0^N r^{1-|\alpha_1|} r^{d-1} dr N^{-|\alpha_2|} \\
&\preccurlyeq \sum_{\alpha_1+\alpha_2=\alpha} N^{-|\alpha_2|} N^{d+1-|\alpha_2|} \\
&= N^{d+1-|\alpha|}.
\end{aligned}$$

Let us now consider two cases: $\alpha = 0$ and $\alpha = d + 2$:

$$\begin{aligned}
\alpha = 0 &\Rightarrow |\nabla \check{m}_N| \leq N^{d+1} \\
\alpha = d + 2 &\Rightarrow |x^{d+2} \nabla \check{m}_N| \leq N^{d+1-d-2} = \frac{1}{N} \\
&\Rightarrow |\nabla \check{m}_N| \preccurlyeq \frac{1}{N} \frac{1}{x^{d+2}}
\end{aligned}$$

Thus:

$$|\nabla \check{m}_N| \leq \min \left(N^{d+1}, \frac{1}{N} \frac{1}{|x|^{d+2}} \right).$$

In conclusion¹⁹:

$$\begin{aligned}
|\nabla K| &\leq \sum_{N \in 2^{\mathbb{Z}}} |\nabla(\check{m}_N)| = \sum_{N \geq \frac{1}{|x|}} |\nabla(\check{m}_N)| + \sum_{N \leq \frac{1}{|x|}} |\nabla(\check{m}_N)| \\
&\leq \sum_{N \geq \frac{1}{|x|}} \frac{1}{N} \frac{1}{|x|^{d+2}} + \sum_{N \leq \frac{1}{|x|}} N^{d+1} \\
&\preccurlyeq \frac{1}{|x|^{d+1}}.
\end{aligned}$$

□

Another classical use of this locator is in the PDEs theory, where it ensures the validity of a chain rule for fractional derivatives.

¹⁹For dyadic sums it is true that $\sum_{N \geq N_0} N = \frac{2}{N_0}$ and $\sum_{N \leq N_0} N = 2N_0$.